Promotions, Unemployment, and Labor Market Sorting Under Asymmetric Information

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Abstract

This paper studies how promotions and unemployment shape labor-market sorting and wages. Using the 1979 National Longitudinal Survey of Youth (NLSY79), I show that promotions amplify the wage losses associated with unemployment spells: previously promoted workers experience larger wage losses upon re-employment, and those losses increase as the spell prolongs, even conditional on pre-displacement wage and tenure. I interpret these facts through a structural model of promotions and job search in which firms learn about worker ability from a worker's history of promotions and unemployment spells. When hiring a worker out of unemployment, firms use this information to assign him a position. Because unemployment duration is a noisy signal of ability, the model implies that unemployment can serve as a shock to labor-market sorting through which firms may either correct an erroneous promotion or incorrectly move capable workers back down the job-ladder. Because UI benefits alter search behavior, they also change how much an unemployment spell reveals about a worker. Relative to a flat benefit profile, UI plans that tie benefits to past wages worsen sorting and lower aggregate output, while policies that target benefits toward low-ability workers improve sorting and raise output – albeit with distributional effects.

JEL Codes: D82, D83, J62, J64, J65, M51
Keywords: Promotions, unemployment duration, asymmetric information, employer learning, labor market sorting, unemployment insurance

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1 Introduction

A central task of the firm is to organize its labor effectively by sorting workers into positions according to ability. One key mechanism firms utilize to sort their existing personnel is the promotion. Over time, a firm observes its worker's performance and uses this information to make a decision on whether or not to move him from a low-ability position into a high-ability position. Not only does this result in an increased wage, but it also provides a signal of the worker's ability to the rest of the labor market. Thus, when hiring new workers, firms may use this signal to assign potential employees into either low-ability or high-ability positions. In this paper, I study the role of promotions, job loss, and unemployment as signals of ability in the firms' assignment decision.

I begin by studying how promotions affect the wage dynamics of workers *after* they separate from the firm that promoted them. Using panel data from the 1979 National Longitudinal Survey of Youth (NLSY79), I find that over 25% of promotions are given to workers who are plausibly low-ability. These individuals are unique among promoted workers in that they are very weakly attached to the labor market and their wages decrease over a 15-year post-promotion window (avg. 25% decline). Furthermore, I find that both entry into unemployment and its duration is particularly costly for workers who were promoted by the employer from which they separated. On average, a promoted worker's first re-employment wage is 15% less than his pre-separation wage, and for each month he remains unemployed, his re-employment wage drops by an additional 0.55 percentage points. For non-promoted workers, these values are merely 1% and 0.14 percentage points per month, respectively.

I use these facts to expand upon existing theory on promotions and the internal labor market and to estimate a model that focuses on the relationship between promotional status and job loss. The key mechanism is that while promotions signal high ability, long unemployment spells are counteracting signals, such that many workers who lose their high-type position are eventually re-employed in the low-type position once again. The fundamentals of the model's internal labor market are slight modifications of the theory presented in Gibbons and Waldman (1999) and Waldman (1984). Workers are endowed with a fixed, unobservable type (high or low), and firms may assign workers freely into either high- or low-type positions. Position types are differentiated only by their production function as it relates to the employed worker. Low-type workers are more productive in low-type positions and high-type workers in high-type positions. To maximize profit, firms attempt to assign high-type workers into high-type positions and vice versa; however, because worker type is not directly observable, they learn over time and promote workers whom they sufficiently believe to be high-types. To promote a worker simply means to move him from a low-type position into a high-type one.

The key features I add are in the spirit of Gibbons and Katz (1991). Formally, I expand upon the aforementioned theory on promotions and the internal labor market to include worker mobility and job-search. While employed, a worker may lose his job

either exogenously due to random chance or endogenously if expected profit for the firm becomes negative. Firms are aware that low-ability workers are more likely to be separated endogenously, so separation serves as a negative signal of ability. Upon separation, the worker enters unemployment and begins job search. Each period, the worker chooses a level of effort to exert, where effort dictates match probability but comes at a utility cost. I assume that high-type workers are more productive in job search such that, given a level of effort, they will match with greater probability than their low-type counterparts. Just as in Gibbons and Katz (1991), firms are asymmetrically informed about a worker's ability. While employing a worker, the firm periodically observes output as a signal and uses this information to update its prior. Firms *not* employing the worker do not observe his performance directly. Instead, they are informed by the worker's employment history, which is the sequence of periods he spent unemployed, employed in a high-type position (employed-H), and employed in a low-type position (employed-L).

With these added features, the model matches the observed relationship between promotions, job loss, and wages. First, promoted individuals face particularly high penalties for job loss because of the signal it sends to outside firms. Due to asymmetric information, when a promoted worker enters unemployment, firms cannot distinguish between an exogenous separation and a firing. Thus, many previously promoted workers are hired into a low-type position at a much lower wage. Second, not only is entry into unemployment a negative signal, but so too is the duration of the unemployment spell. Each period, a high-type worker will match with a greater probability than a low-type worker. Because of this, each continued period of unemployment sends a negative signal to prospective firms. As before, this is of particular consequence for workers who were previously promoted because it further decreases the likelihood that they retain their high-type position at the new match.

Within the model, unemployment serves as an opportunity for firms to reassign workers to positions. That is, when hiring a worker, the firm's objective is to assign him to a low-type position if he is low-type and to a high-type position otherwise. To evaluate sorting in the model, I compute the rate at which firms incorrectly assign workers into positions upon hiring them. Error-H is the rate at which firms reassign high-type workers to low-type positions conditional on them having left a high-type position. Error-L is the rate at which firms assign a low-type workers into high-type positions conditional on them having left a high-type position. Both error rates condition on the worker having left a high-type position. This is because workers almost never move from low- to high-type positions after an unemployment spell, so error-H and error-L alone provide sufficient information to evaluate sorting in the model.

Using the model, I show that the design of unemployment policy can have substantial impacts on sorting, output, and wages. Policies that provide greater benefits to unemployed low-type workers or lower benefits to unemployed high-type workers change the effect of unemployment duration as a signal. This is because such policies provide greater (lesser) incentives for high-type (low-type) workers to search, thereby increasing the correlation

between worker type and unemployment duration. I explore two separate policies to target this mechanism. First, I show that UI regimes that pay benefits proportional to a worker's past wage result in worse sorting outcomes and subsequently reduce output and wages. Under proportional UI, error-H rises to nearly 34%, while error-L improves marginally to 17.57%, resulting in a decrease in aggregate output of 0.27% and a reduction in average wages of 0.5%. On the other hand, UI policy can be designed to target low-type workers. Because high-type workers expect shorter unemployment spells, conditioning benefits on an upfront, fixed payment can incentivize high-type workers to opt out. This change increases the potency of unemployment as a signal and reduces error-H to 22% at the expense of increasing error-L to 32%. This change results in an increase in output of 2.5% and an increase in wages of around 4.5%.

1.1 Related Literature

This paper builds on several strands of work in labor economics and macroeconomics. The starting point is the large literature on promotions and internal labor markets, which emphasizes how firms use job assignment and wage growth to organize workers according to ability (Baker et al., 1994a,b; Lazear and Rosen, 1981). Gibbons and Waldman (1999) provide an influential model in which promotions and wage dynamics arise jointly from employer learning and the assignment of workers to positions with different sensitivity to ability, and this framework helps explain the life-cycle decline in promotion rates and the acceleration of wages around promotion. My empirical contribution is to push beyond the within-firm lens by tracing how promotions shape a worker's career *after* he separates from the promoting firm—a dimension that standard internal-labor-market models are not designed to address—and to connect these dynamics to the literature on frictional labor markets and job ladders (e.g. Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002; Moscarini and Postel-Vinay, 2012), where wage growth over the life cycle is tightly linked to job-to-job mobility.

The focus on cross-firm career dynamics connects naturally to the literature on labor market sorting, which studies how heterogeneous workers are assigned to heterogeneous positions. Jovanovic (1979) established that productive match quality is revealed gradually over time and that separation and re-matching serve as mechanisms for moving workers toward better-suited positions. Modern quantitative work such as Lise and Postel-Vinay (2020) extends this to multidimensional worker and firm types and highlights that the quality of sorting has first-order consequences for aggregate output and wages. A central theme in this literature is that frictional search impedes sorting and that improving allocative efficiency requires good information about worker quality. This paper contributes a novel dimension of that problem: promotions shape the informational environment that future employers use when assigning workers, and unemployment spells are the primary occasions on which re-assignment takes place.

This sorting-through-information mechanism links the paper to work on how markets

learn about worker ability and how observable career histories serve as signals. Farber and Gibbons (1996) and Altonji and Pierret (2001) show that wages and employment histories aggregate information about productivity, with the pace of learning identifiable from how the returns to observable skills evolve over experience. A closely related contribution is Gibbons and Katz (1991), who show that the type of separation matters: workers laid off from shrinking firms are treated as lemons by prospective employers and suffer larger subsequent wage losses than workers who separate for other reasons, consistent with layoffs serving as a negative signal about ability. My model extends this logic by interacting the signal from a prior promotion with the signal from a subsequent unemployment spell. Because unemployment puts at risk the discrete wage-gains provided by the high-position, the negative impact of unemployment becomes especially large. This mechanism is related to the classic signaling model of Spence (1973) and to work on statistical discrimination and inference from employment histories (Waldman, 1984; Greenwald, 1986), but with the key difference that the signal is generated endogenously through internal firm decisions rather than through an upfront investment by the worker.

The final strand concerns job separations, unemployment duration, and subsequent earnings. A foundational empirical finding is that job loss generates large and persistent earnings declines, with substantial heterogeneity across workers and separation types (Jacobson et al., 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Schmieder et al., 2016). Theoretically, unemployment duration is itself informative: if high-ability workers find jobs faster, each additional period of non-employment updates a prospective employer's beliefs downward (Lockwood, 1991; Kroft et al., 2013). My results speak to both of these themes. Previously promoted workers experience wage losses after unemployment that far exceed those of the non-promoted, and these losses grow strongly with unemployment duration—a pattern that is difficult to reconcile with pure human-capital depreciation but that follows naturally from the inference mechanism in the model. Taken together, these strands motivate a framework in which promotions are publicly observed, endogenously generated signals, unemployment spells are the occasions on which the market revises those signals, and the design of unemployment insurance shapes the informativeness of duration and therefore the quality of aggregate sorting.

2 Data and Stylized Facts

The primary data source is the National Longitudinal Survey of Youth (NLSY). The NLSY is currently split into two single-cohort panels: the 1979 survey (NLSY79) and the 1997 survey (NLSY97). I utilize only the 1979 survey, as the latter does not collect information on promotions. I then provide several basic facts about promotions in the data, several of which are used to estimate the model, before presenting the key empirical facts regarding promotions, unemployment, and labor-market sorting.

2.1 NLSY79

The 1979 National Longitudinal Survey of Youth (NLSY79) is a longitudinal survey of individuals aged 15–21 on January 1, 1979. Respondents were surveyed annually from 1979 to 1994 and biennially from 1996 to the present. Each survey consists of both a retrospective weekly employment diary and a survey of demographics, wealth, etc. For tractability, both in the empirical analysis and in the subsequent model, I aggregate the data to a monthly frequency when developing the full panel.

Pivotal to this analysis, from 1988-1990 and to 1996-2016, the employment diary includes the respondent's account of his or her promotion history. More specifically, each respondent is asked whether "[he/she] experienced an increase in responsibility in her job". If so, the interviewer follows up by asking, "does [he/she] consider that a promotion". Because of this, I find that the NLSY's definition of a promotion is related to that of Bayer and Kuhn (2023), who define a promotion as an increase in autonomy and responsibility.

Occupation reports are of some concern in the labor literature. Kambourov and Manovskii (2013, 2008) find evidence that the PSID, CPS, and related employment panels over report occupation changes due to issues with measurement and data collection. I, too, find this to be the case in the NLSY. To gather occupation information, NSLY surveyors are direct to assign a 3-digit Census occupation code based on the respondent's description of his or her day-to-day activities. Thus, due to differing interpretations between waves, a worker's occupation may artificially change erroneously. Kambourov and Manovskii (2013) remedy this concern by assigning the modal occupation to each worker's employer spell. The NLSY's job-reporting structure allows me to relax this restriction by exploiting the timing of reports and their use of multi-job spells. During each interview, respondents may report up to five separate jobs to the same surveyor along with the dates they were employed. Occasionally, a respondent reports working two separate jobs with the same employer in a given interview. Because the surveyor is fixed for a given interview, this reflects an intentional separate report by the respondent and *not* a change in interpretation. Thus, respondents in my panel may change occupations within the same employer only if the change is reported within an interview-block; otherwise I default to the employer-spell classification used in Kambourov and Manovskii (2013).

Sample Selection To construct my sample from the NLSY79, I first retain only male respondents with valid demographic information. I then drop all observations with missing employer, occupation, wage, or hours-worked information and all individuals with missing demographic information. Since promotion information is available only for 1988–1990 and 1996–2016, I drop observations with missing promotion information if they fall within those years and then drop observations with other missing information, such as test scores. Similar to Guvenen et al. (2020), I drop observations for each individual before the individual enters the labor market for the first time and before the individual leaves school.

Finally, I drop all observations for individuals with response rates below 85%[1]. Table 2.1 provides details on this selection process. The final panel consists of 2,193 individuals and 1,899,411 person-month observations.

Selection Criteria	**Individuals**	**Observations**
	12,686	6,698,208
Only Males	6,403	3,380,784
Missing Demographics	6,393	3,375,504
Missing Employment	6,314	2,234,942
Missing Promotions	6,314	2,234,238
Other Missing	5,260	1,899,411
Post-School	4,568	1,408,944
Post Labor Market Entry	3,199	1,232,946
Response Rate $\geq$ 85%	2,193	966,802

Notes: Demographics include race and education information. Missing employment means that either occupation or employer codes are missing. Missing promotion observations are dropped only for 1988–1990 and 1996–2016 because promotions are reported only for those years. Other missing information includes missing test scores, missing wages, or wage discrepancies across various responses.

Table 2.1: Sample Selection

2.2 Characteristics of Promotions

Promotions are frequent early in the life cycle and decline sharply with age. Panel A of Figure 2.1 plots the monthly promotion rate by age. At the start of a career, the average worker is promoted with probability $\sim$ 2% per month (20% per year). By age 35, the rate falls to under 1% per month (10% per year) and approaches zero near the end of the career[2]. Panels B and C show analogous declines in occupational and employer churn, respectively.

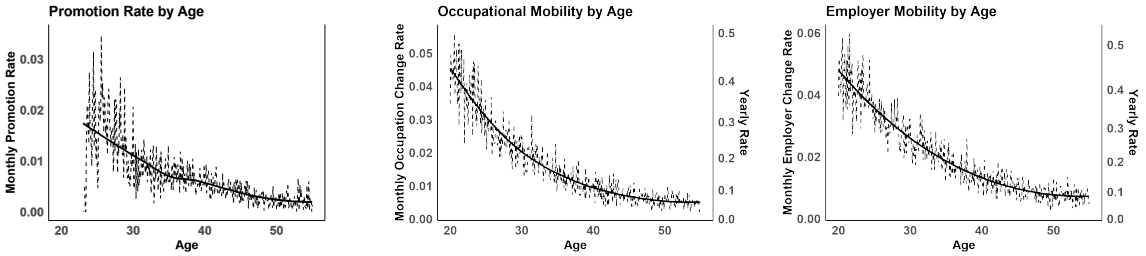
A natural interpretation of these profiles is learning: over time, workers and firms acquire information about match quality and comparative advantage, reducing the need for reassignment. Gervais et al. (2016) and Papageorgiou (2014) argue that this decline in occupational churn reflects workers becoming progressively better informed about which occupations suit their skills[3].

Unlike occupation and employer changes, promotions are absent for a sizable minority of workers. In the NLSY, only 4.17% and 2.73% of workers go through their entire careers

[1] Changing the response-rate restriction has little effect on the empirical results discussed in Section 2.

[2] This trend holds across the education gradient, though more highly educated workers are positively shifted relative to those with less education.

[3] Guvenen et al. (2020) and Lise and Postel-Vinay (2020) describe similar environments. Guvenen specifically finds evidence that worker churn increases as the distance between the vector of workers' skills and employers' requirements increases.



(a) Promotion (b) Occupation (c) Employer

Figure 2.1: Life-Cycle of Employment Mobility

Notes: Panels A–C provide the monthly rates of promotions, occupation changes, and employer changes for each age (by month). Dashed lines represent raw data. Solid lines represent local-linear smoothing. A transformation of the monthly rate is shown along the right-hand side for interpretability.

without changing employers or occupations, respectively, and the average worker changes occupations nearly 6 times and employers 7.6 times. In contrast, just over 30% of individuals never receive a promotion, and nearly 30% receive only one. Figure 2.2 reports the full distribution.

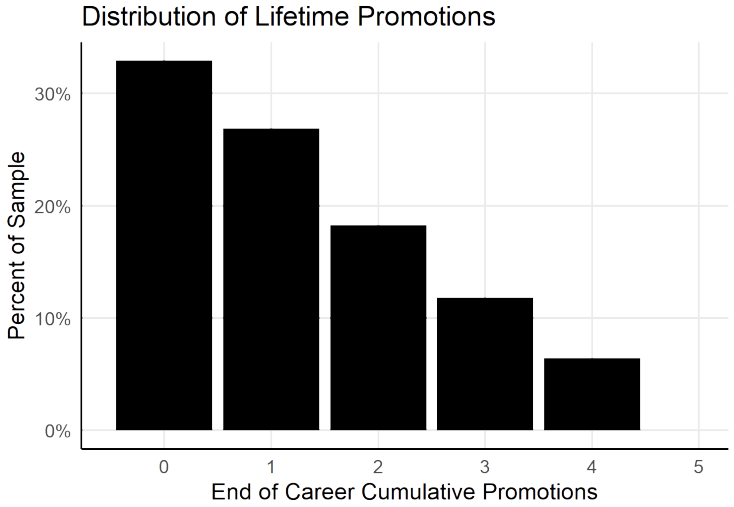
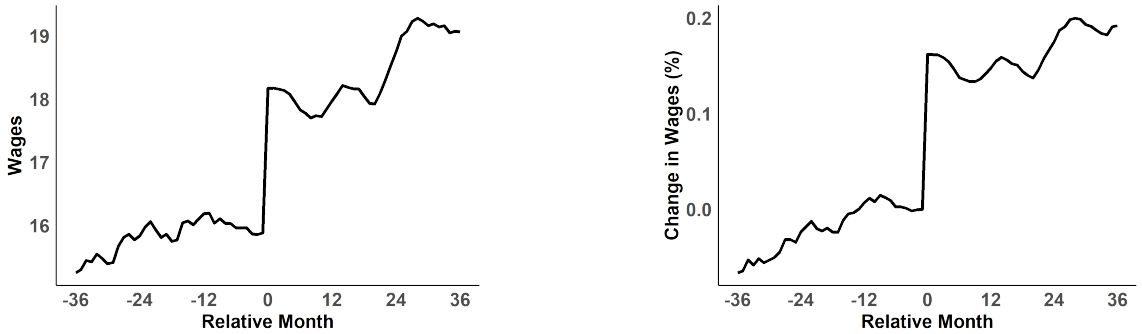


Figure 2.2: Distribution of Life-Time Promotions

Notes: The figure shows the percentage of the NLSY sample who receive 0–5 promotions over the course of their careers. Those who are promoted 6 or more times, of whom there are relatively few, are omitted.

Turning to wages, promotions are associated with large contemporaneous gains. The average employee's first promotion raises current wages by 17%, about \$2.4/hour. Previous empirical studies of promotions (Baker et al., 1994a,b; Gibbons and Waldman, 1999) find that wage growth often accelerates after promotion relative to the pre-promotion trend.

The NLSY sample shows the same pattern, though more modestly: in the 36 months leading into a worker's promotion, wages rise on average from \$15.1 to \$18.9 (\$0.4/year); in the 36 months following the promotion, wages rise from \$18.2 to \$19.2 (\$0.5/year). Figure 2.3 summarizes these dynamics: Panel A plots wage levels around the first promotion and Panel B reports percent changes relative to the last pre-promotion wage.



(a) Hourly Wages

(b) Wage Changes (%)

Figure 2.3: Wage Effect of Promotions

Notes: Average wages are plotted on an event horizon centered at a worker's first promotion. Panel A shows hourly wages, while Panel B shows log wages normalized to event time $t = -1$.

Despite the large average effect, there is a large degree of heterogeneity in the effect of promotions on wages. Table 2.2 reports selected quantiles of the wage change at promotion. The lower tail is particularly striking: nearly 20% of promotions have a zero or *negative* contemporaneous wage effect, and the 10th percentile experiences a wage decline exceeding 10%. At the upper tail, the 90th percentile gains about 50%.

Gibbons and Waldman (1999) argues that getting promoted is a job assignment into a different production function. Firms promote workers as their human capital grows beyond the threshold at which production in a promoted position exceeds production in the non-promoted position. Assuming that wages track productivity in some way, dispersion such as that in Table 2.2 suggests that motives other than production affect both an employer's incentive to offer a promotion and a worker's incentive to accept it. For simplicity in the model, I later load the entire effect onto taste shocks on the part of the employer.

Table 2.2: Distribution of Promotions

	p10	p20	p50	p80	p90
Wage Growth (%)	-11.74	0	13.39	33.66	49.79

Notes: Values represent the 10th, 20th, 50th, 80th, and 90th percentiles of promotion-induced wage gains.

2.3 Promotions and Unemployment

As documented in the previous section, promotions are often associated with large wage gains averaging over 15%. However, the lasting effects of a promotion are highly variable. I exploit the panel nature of the NLSY to follow workers over a 15-year window following their promotion. In doing this, I identify a group of promoted individuals whom I refer to as "plausibly low-ability." The reason for this designation is that, compared with the rest of the promoted subgroup, these workers are particularly weakly attached to employment and experience very negative wage outcomes.

Figure 2.4 plots the average path of wages for two subgroups: the dotted red line plots average wages for workers who experience above-median exposure to unemployment in the post-promotion period, while the solid blue line plots the same for workers who experience below-median exposure. "Exposure" is determined by the number of months a worker is not employed *after* he receives a promotion. The trend is clear: starting nearly 3 years after promotion, the wages of high-exposure workers begin to deviate from those of low-exposure workers. By 15 years, their wages have decreased by an average of 25%—10% more than the average wage gain from their promotion. This drastic effect of unemployment is not shared by workers without promotions. To compare the post-promotion path of of wages to those that never receive a promotion, I create "synthetic" promotions for the never-promoted by randomly pulling a date for each worker from the distribution of first-promotions in the ever-promoted group. In the 15-years following these "synthetic" promotions, workers who are weakly attached to the labor force experience a wage *gain* of nearly 5% and the wages for those who are strongly attached evolve nearly in parallel to the wages of strongly attached promoted workers.

I find that this deviation in wage profiles is primarily explained by the dynamics of wages around unemployment spells. Three facts stand out: (i) after losing a job, promoted individuals experience wage declines of $\sim 14\%$, whereas the effect of job loss for the non-promoted is close to zero; (ii) employer and occupation transitions that do *not* involve an intermittent unemployment spell affect the promoted and the non-promoted in a similar, positive way; and (iii) the effect of unemployment *duration* is nearly 0 for non-promoted individuals, but duration rapidly diminishes wages for those who were promoted. The remainder of this section documents and interprets these facts.

I begin by studying how the consequences of job loss vary with prior promotion status. A large existing literature (Jacobson et al., 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Schmieder et al., 2016)[4] studies separations from employment and their effects on worker outcomes. The consensus is that job loss has an immediate negative effect on wages, and losses grow as workers remain out of employment.

Separating the sample into promoted and non-promoted workers, the average wage decline through an unemployment spell is roughly 14% for the promoted and nearly 0% for the non-promoted. Figure 2.5 plots log wages, normalized to the last wage before an

[4] this is not an exhaustive list

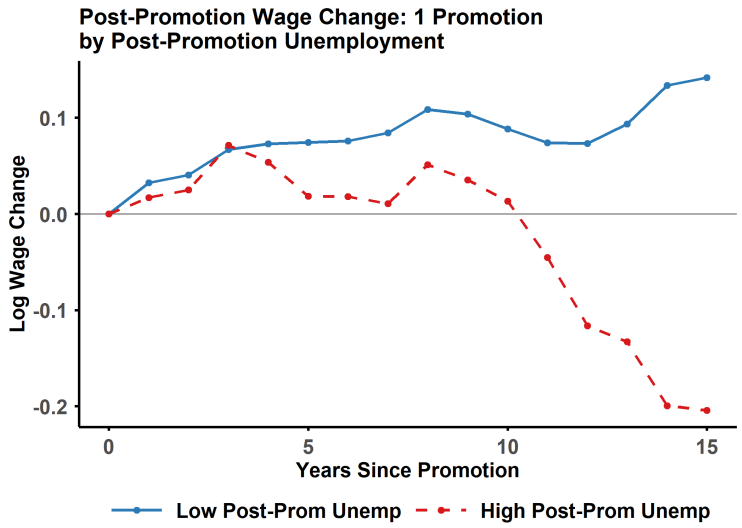


Figure 2.4: Post-Promotion Wage Paths by Unemployment Exposure

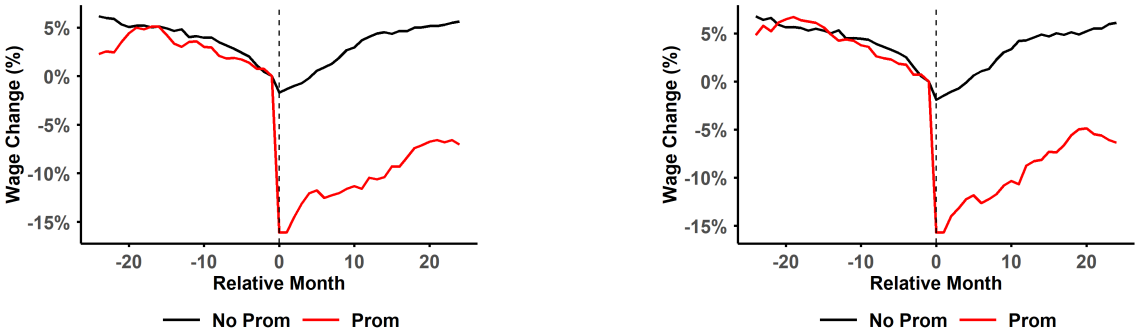
Notes: Figure plots average log-wages, normalized to 0 at the time of a worker's first and only promotion. The dotted red line represents individuals that experience above-median exposure to unemployment, measured in cumulative months over the 15-years following the promotion, while the solid blue line represents workers that experience below median exposure to unemployment.

unemployment spell, over a 48-month window (24 months before job loss and 24 months after re-entry). Panel A includes all unemployment-to-employment (U2E) transitions, while panel B restricts to U2E transitions that also involve an occupation change[5].

Motivated by this descriptive evidence, I estimate equation 2.1 using a stacked event-study design. For each worker i and unemployment spell s, I keep only observations in the 49-month window centered on the spell, i.e., relative months $r \in \{-24, \ldots, 24\}$. I then stack these spell-specific windows into one estimation sample. If two unemployment spells are close enough that their windows overlap, the same worker-month can appear more than once in the stacked sample, once for each relevant spell window. Promoted workers are typically more tenured, more educated, and employed in more demanding occupations; failing to account for these differences would mechanically inflate the estimated role of promotion status.

$$\Delta \log w_{isr} = \sum_{\substack{k=-24 \\ k\neq -1}}^{24} \beta_k \mathbf{1}\{r = k\} \, \text{Prom}_{is,-1} + \sum_{\substack{k=-24 \\ k\neq -1}}^{24} \gamma_k \mathbf{1}\{r = k\} + \alpha_{is} + \lambda_{c(isr)} + \beta_{\mathbb{X}} \mathbb{X}_{isr} + \varepsilon_{isr} \tag{2.1}$$

[5]Occasionally, workers lose their job and return to work with the same employer. These cases are not included in either panel



(a) Employer Changes

(b) Occupation Changes

Figure 2.5: Wages Around U2E Transitions

Notes: Both panels plot average log wages normalized to a worker's last pre-unemployment wage, $t = -1$. Observations to the left of the dotted line are from before an unemployment spell began, whereas observations to the right are from after it began. Thus, for each worker, there is a varying amount of time between observations $t = -1$ and $t = 0$. The sample for Panel A consists of all unemployment spells that satisfy the following conditions: (i) they occur ≥ 24 months after a worker enters the panel, (ii) they occur ≥ 24 months before he leaves the panel, and (iii) they result in an employer change. Panel B's sample satisfies the same conditions and additionally requires an occupation change.

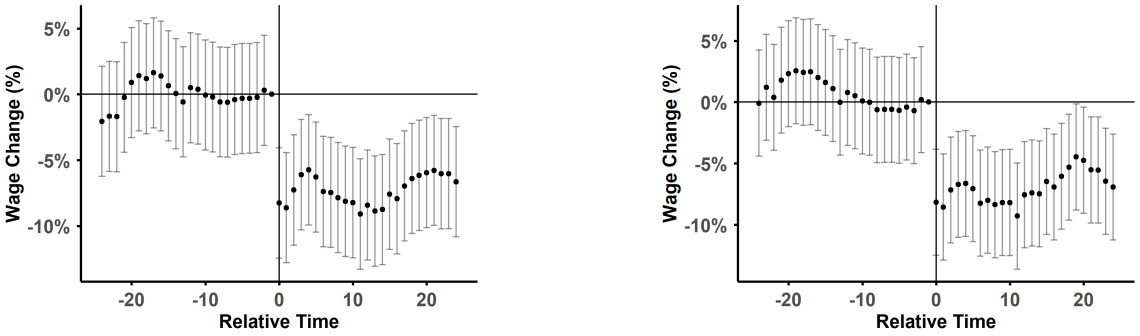
$$\text{where, } \Delta \log w_{isr} \equiv \log w_{isr} - \log w_{is,-1},$$

In it, i indexes workers, s indexes unemployment spells, and r indexes relative month within spell s, where $r = -1$ is the last month before entry into unemployment. The indicator $\text{Prom}_{is,-1}$ equals one if worker i was in a promoted position at relative month -1 for spell s, and zero otherwise. Included in $\mathbb{X}_{isr}$ is a third-order polynomial in age, employer tenure, occupational tenure, continuous employment tenure, and experience. $\mathbb{X}_{isr}$ also includes indicators for the 2-digit SOC occupation code and linear terms in both unemployment duration and occupational cognitive requirements as defined by Guvenen et al. (2020). The term α_{is} contains worker $\times$ unemployment-spell fixed effects[6], and $\lambda_{c(isr)}$ is a calendar-time fixed effect for the underlying month-year of observation.

The estimates of β_k are reported in Figure 2.6. As expected, controlling for loss of tenure and occupational "hierarchy" attenuates the gap, but the overall effect remains sizable: promoted individuals lose uniquely more wages ($\sim 7.5\%$ more) than do their non-promoted counterparts. As in Figure 2.5, I report results for all U2E employer changes (panel A), and for the sub-sample that also involves an occupation change (panel B). The results are nearly identical across the specifications.

When transitions do not involve unemployment, differences by promotion status are less pronounced. On average, an employer-to-employer (E2E) transition is associated with nearly a 5% wage gain for promoted workers and around a 12% gain for non-promoted workers. The gap (about $\sim 7\%$) is roughly half the size of the gap associated with un-

[6]Specifications with only individual fixed effects do not change the results

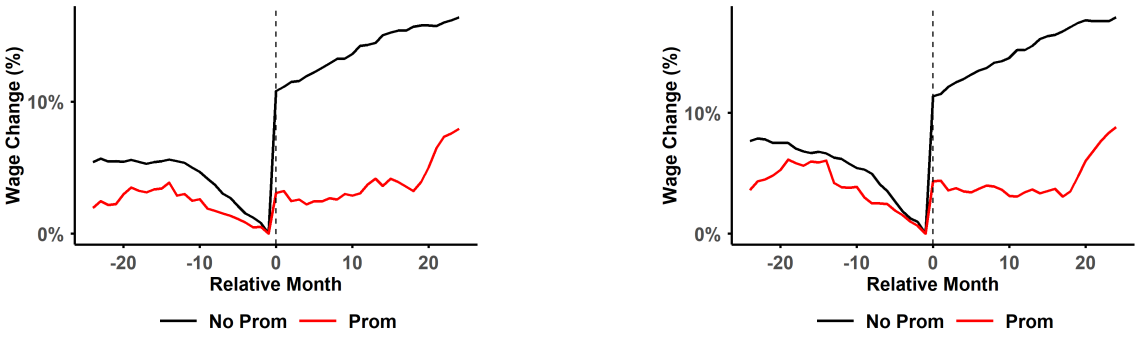


(a) Employer Changes (b) Occupation Changes

Figure 2.6: U2E Event Study - Effect of Promotions

Notes: Plotted are the Promotion × relative-month fixed effects around an unemployment spell. Significant negative coefficients for $t \geq 0$ indicate that promoted individuals lose more wages in unemployment than the non-promoted. Regressions control for third-order polynomials in tenure, age, and experience; indicators for 2-digit SOC codes, and a linear term in both unemployment duration and occupation cognitive requirements (Guvenen et al., 2020). Panels A and B are separated as described in Figure 2.5.

employment transitions (about $\sim 14\%$), and both groups experience positive wage gains. Figure 2.7 replicates the event-time plot in Figure 2.5, centered at an E2E transition.



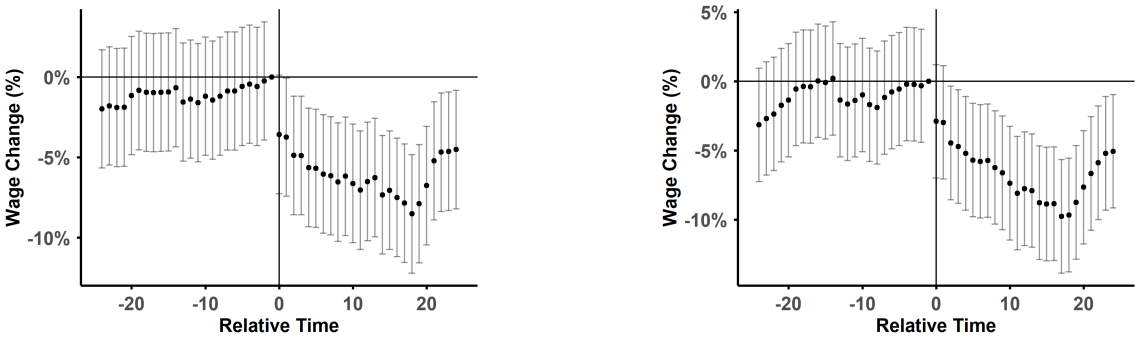
(a) Employer Changes (b) Occupation Changes

Figure 2.7: Wages Around E2E Transitions

Notes: Panels A and B both plot average log wages, normalized to time $t = -1$, in a 24-month window centered around a worker's employer-to-employer transition. Panel B plots a subset of workers in Panel A whose employer change also coincided with an occupational change.

I estimate a counterpart to equation 2.1, where s instead indexes a worker's s^{th} E2E transition. Figure 2.8 reports the resulting set of β_k. Conditional on observables and fixed effects, promoted and non-promoted workers exhibit broadly similar immediate wage dynamics around employer changes *without* intervening unemployment. Notably, however, wage growth for the non-promoted group outpaces that of promoted workers in the subsequent two years. Consistent with upward job ladders, I find that the non-promoted group's

promotion rate rises over this horizon, suggesting that some E2E moves reflect search for positions with higher promotion prospects.



(a) Employer Changes

(b) Occupation Changes

Figure 2.8: E2E Event Study - Effect of Promotions

Notes: Plotted are the Promotion × relative-month fixed effects around an employer-to-employer (E2E) transition. Panels A and B are separated as described in Figure 2.7. The specification is identical to that in Figure 2.6, where the only difference is the center of the horizon.

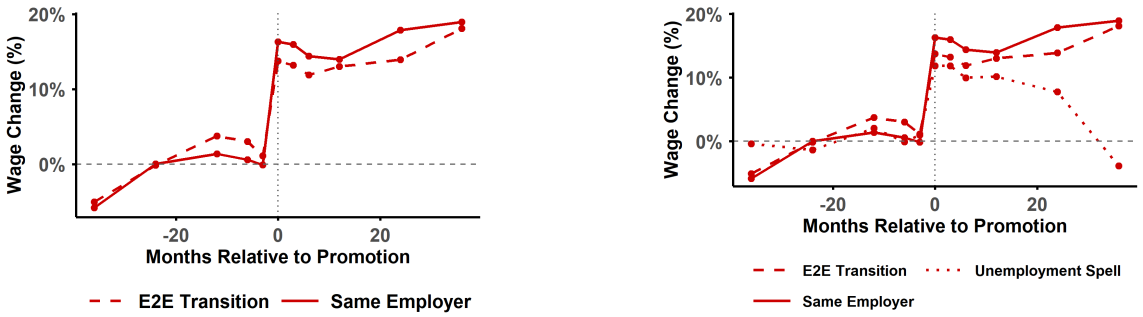
This result is corroborated when comparing the wages of promoted stayers relative to promoted workers that experience E2E transitions. Figure 2.9 separates promoted workers into two groups: those that change employers in the 24-month window following a promotion and those that stay. Panel (b) adds in a third group of promoted workers that become unemployed and re-employed during the same 24-month window. The result is striking: among promoted workers, the path of wages for those who stay with their employer evolves nearly parallel to those that change employers through the E2E channel. Panel (b) shows very clearly that the same is not true for those that separate from employment.

Finally, I study heterogeneity in job-loss costs by unemployment *duration*. Unlike non-promoted individuals, the wage-loss function for promoted workers is strongly duration-dependent. Each additional month of unemployment reduces the next wage of a promoted worker by nearly 0.97 percentage points on average, compared with roughly 0.43 percentage points for the non-promoted. Figure 2.10 provides a visual summary of this pattern.

A promoted worker who enters unemployment and re-enters very quickly experiences an average loss of about 7.5% of his previous wage, while a similar non-promoted individual sees no change. The gap more than doubles by 18 months, the 80th percentile of duration in the NLSY[7].

I estimate a flexible version of the wage-loss function in equation 2.2, where i indexes individuals and s indexes a worker's unemployment spells, i.e., $s = 1$ denotes a worker's first unemployment spell, $s = 2$ denotes his second, and so on. $\Delta \log w_{is}$ is the difference

[7]Recall that duration is not limited to months the CPS considers unemployed. In fact, a majority of *non-employment* in the NLSY is not reported as unemployment.



(a) Employer Changes

(b) Occupation Changes

Figure 2.9: Post-Promotion Wage Paths by Subsequent Employment Transition

Notes: Panel (a) plots average log-wages relative to pre-promotion wages for workers in 2 categories: the dashed line represents wages for workers who are promoted and subsequently change employers (E2E) in the 24-month post-promotion window, while the solid line represents wages for workers who are promoted and remain with their employer for the duration of the post-promotion window. Panel (b) adds those who experience a full U2E change within the post-promotion window (job loss and new match).

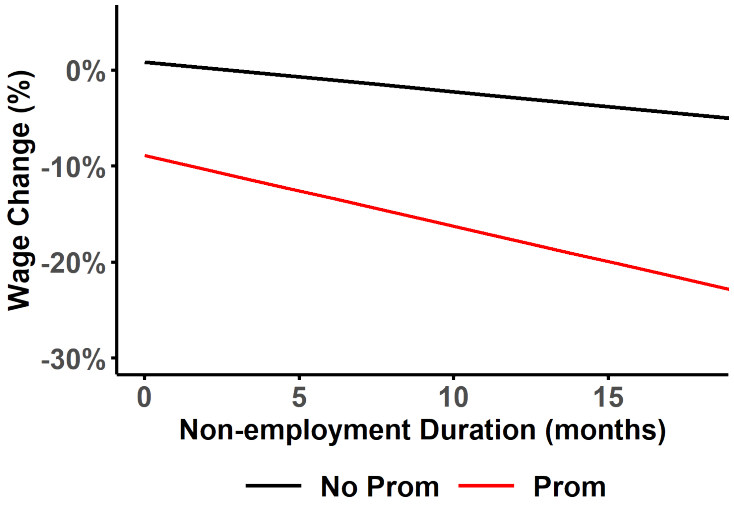


Figure 2.10: Wage Loss as a Function of Duration and Promotion

Notes: The figure plots an estimated linear trend of wage loss as a function of promotion status before job loss and the duration of the unemployment spell.

in log wages immediately before and after worker i's s^{th} unemployment spell. $Prom_{is}$ is an indicator equal to one if worker i received a promotion from the employer that immediately preceded his s^{th} unemployment spell, and zero otherwise. The vector $\mathbb{X}_{is}$ contains a worker's experience, age, and tenure immediately before the sth unemployment spell, as well as Req_Rank, the difference in occupational skill requirements (Guvenen et al., 2020) between his pre- and post-unemployment occupations.

$$\Delta \log w_{is} = \alpha + \beta_1 Prom_{is} + \beta_2 U_{is} + \beta_{\mathbb{X}} \mathbb{X}_{is} + \beta_3 Prom_{is} U_{is} \tag{2.2}$$

The results from this specification are provided in Table 2.3.[8] The estimates mirror Figure 2.10: promoted individuals are uniquely harmed as an unemployment spell is prolonged (around 0.7 p.p. per month), whereas non-promoted workers exhibit a much smaller negative effect (around 0.2 p.p. per month). Column (1) reports the baseline specification, and column (2) restricts the sample to unemployment spells that also involve an occupational change.

3 Model

Using the relationship between promotions, employer changes, unemployment, and wages, I estimate a structural model of promotions and employer mobility in which both entry into unemployment and the duration of the spell serve as firms' sources of information in positional assignment. The rest of the section lays out the model. Section 3.1 describes the basic environment and worker behavior, Section 3.2 describes firms and their beliefs, and Section 3.3 defines the equilibrium.

3.1 Workers

There is a continuum of heterogeneous workers. A worker's type $z \in \{0, 1\}$ is fixed over the life cycle, where $z = 1$ (high-type) workers are more productive both at work and in job search. Type is privately known to the worker and imperfectly observed by employers; thus, employers form expectations about a worker's type.

Beliefs are denoted by π and χ, where $\pi = P_j(z_i = 1)$ and $\chi = P_{-j}(z_i = 1)$. Here i indexes the worker, j is the worker's *current* employer, and $-j$ denotes all other (non-matched) employers. Omitted from notation is the set of information firms use to develop their priors, χ and π. This is detailed formally in Section 3.2.

Each period, workers exit the model with probability d and are replaced by new entrants whose type is drawn from the unconditional distribution Π_z.

[8] Omitted only are the estimated occupation fixed-effects

Table 2.3

	Dependent variable:	
	(1)	(2)
Prom	−0.011	−0.024
	(0.024)	(0.026)
UnemCount	−0.002***	−0.001**
	(0.001)	(0.001)
Exp	−0.013	−0.027
	(0.014)	(0.016)
Exp^2	0.0003	0.001
	(0.001)	(0.001)
Exp^3	0.00000	−0.00002
	(0.00002)	(0.00002)
Age	0.015	0.062
	(0.052)	(0.060)
Age^2	0.0002	−0.001
	(0.001)	(0.002)
Age^3	−0.00000	0.00001
	(0.00001)	(0.00001)
Tenure	−0.029***	−0.021**
	(0.007)	(0.009)
$Tenure^2$	0.001	0.001
	(0.001)	(0.001)
$Tenure^3$	−0.00002	−0.00001
	(0.00002)	(0.00003)
Req_Rank	0.089***	0.095***
	(0.028)	(0.029)
Prom:UnemCount	−0.005***	−0.004***
	(0.001)	(0.001)
Constant	−0.309	−0.853
	(0.629)	(0.719)
Observations	5,406	4,147
R^2	0.046	0.053
Adjusted R^2	0.040	0.045
Residual Std. Error	255.882 (df = 5371)	261.593 (df = 4112)
F Statistic	7.698*** (df = 34; 5371)	6.757*** (df = 34; 4112)

*p<0.1; **p<0.05; ***p<0.01

Notes: (1) regresses the change in log wage for individuals who moved into unemployment. (2) does the same, but restricts the sample to those that also changed occupations. Req_Rank is the difference in percentile of the post-unemployment occupation from the pre-unemployment occupation.

Employment Workers are either employed ($e = 1$) or unemployed ($e = 0$). Conditional on employment, the matched employer assigns a promotion status $p \in \{0, 1\}$, where $p = 0$ denotes a non-promoted position and $p = 1$ denotes a promoted position. While employed, workers are paid a wage $w(\pi, p) = \omega \mathbb{E}[y(z, p) - k_p|\pi]$, where $y(z, p)$ is production, $\omega \in [0, 1]$ is the wage rate, and k_p is a fixed cost of maintaining the match[9]. Production is given by equation 3.1. Here, $\varepsilon \sim N(0, \sigma_y^2)$ is white noise, and c_p is the penalty from assigning a worker to an ill-suited promotion status.

$$\log(y) = z - c_p|p - z| + \varepsilon \tag{3.1}$$

Equation 3.1 implies complementarity between promotions and high types or quality matches: misassignment reduces output through $c_p|p - z|$. Low-type workers are less productive in promoted positions, and high-type workers are less productive when not promoted. Empirically, Guvenen et al. (2020) and Lise and Postel-Vinay (2020) document sizable production losses from mismatches between worker skills and job assignments. The intuition is that low-type workers do not have the necessary skills and that high-type workers are not maximizing their potential without the added responsibilities.

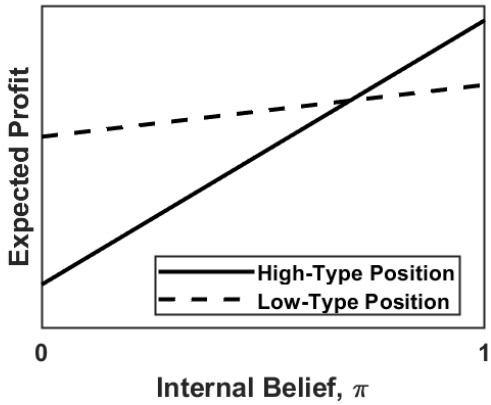
The specification is consistent with standard promotion models. In Gibbons and Waldman (1999), production is linear in general human capital, and the assumptions $y_0(0) > y_1(0)$ and $y_1'(h) > y_0'(h)\ \forall\ h$ deliver a single-crossing condition: the employer promotes when human capital (h) exceeds a threshold $\bar{h}$ satisfying $\bar{h} := y_0(\bar{h}) = y_1(\bar{h})$. Under equation 3.1 and the wage rule, the employer's promotion decision similarly exhibits single crossing in π, as illustrated in Figure 3.1.

At the end of each period, an employed worker separates exogenously into unemployment with probability δ. Conditional on no separation, the worker is reviewed with probability ρ. A review triggers Bayesian updating of the matched employer's belief π and a subsequent decision to fire, promote, or retain the worker in the current position. I rule out demotions, both because they are rare in the data ($< 1\%$ per year) and because the restriction substantially simplifies computation.

The value functions for an employed non-promoted worker and an employed promoted worker are provided in equations 3.2 and 3.3, respectively. In each, $\mathcal{S}$ is the set of information that describes the worker's employment spell. Specifically, it contains the belief with which the worker began employment, the number of periods the worker has been employed, and, if applicable, the number of periods the worker has been in a promoted position. Both π' and $\mathcal{S}'$ represent the evolution of the state variables. For $\mathcal{S}$, this is an additional period of tenure and, when used in V_e^P, an additional period of promotional tenure. π' is shorthand for the new belief conditional on a review. It is detailed formally in equation 3.5. The function $\Gamma(\mathcal{S})$ is the belief transition function for the external market and is defined in Section 3.2. The value for an unemployed worker is given by V_u.

[9] One interpretation of k_p is rent or opportunity cost.

Figure 3.1: Expected Profit as Function of π, p



Notes: The employer's expected profit as a function of its belief about its worker, π, and the promotion status it assigns to him. The intersection, $\bar{\pi}$, is not guaranteed to be in $[0,1]$, but in estimation, $\bar{\pi}$ will be used to target the total promotion rate in first jobs.

$$V_e^{NP}(z,\pi,\mathcal{S}) = w(\pi,0) + \beta(1-d)\begin{pmatrix} \delta V_u(z,\Gamma(\mathcal{S}')) \\ +\,(1-\delta)\rho\,\mathbb{E}\big[F(\pi',0)V_u(z,\Gamma(\mathcal{S}'))\big] \\ +\,(1-\delta)(1-\rho)\,V_e^{NP}(z,\pi,\mathcal{S}') \\ +\,(1-\delta)\rho\,\mathbb{E}\big[(1-F(\pi',0))(1-Pr(\pi'))V_e^{NP}(z,\pi',\mathcal{S}')\big] \\ +\,(1-\delta)\rho\,\mathbb{E}\big[(1-F(\pi',0))Pr(\pi')V_e^{P}(z,\pi',\mathcal{S}')\big] \end{pmatrix} \tag{3.2}$$

$$V_e^{P}(z,\pi,\mathcal{S}) = w(\pi,1) + \beta(1-d)\begin{pmatrix} \delta V_u(z,\Gamma(\mathcal{S}')) \\ +\,(1-\delta)\rho\,\mathbb{E}\big[F(\pi',1)V_u(z,\Gamma(\mathcal{S}'))\big] \\ +\,(1-\delta)(1-\rho)\,V_e^{P}(z,\pi,\mathcal{S}') \\ +\,(1-\delta)\rho\,\mathbb{E}\big[(1-F(\pi',1))V_e^{P}(z,\pi',\mathcal{S}')\big] \end{pmatrix} \tag{3.3}$$

Unemployment While unemployed, workers consume home production b and choose search effort $\theta \in [0,\infty)$ to maximize the present value of expected utility. Search effort maps into a job-finding probability via $p(\theta,z) = 1 - e^{-\alpha_z\theta}$, where $\alpha_1 > \alpha_0$, so high types convert effort into offers more effectively. Effort entails utility cost $\nu(\theta) = \xi\theta^{\gamma}$, where $\xi > 0, \gamma > 1$.

Equation 3.4 gives the unemployed value function for a type-z worker facing market belief χ. The objects V_P and V_{NP} denote the values upon entering employment in a promoted and non-promoted position, respectively, conditional on type z, internal belief

π, and external belief χ. Workers who become employed do so with the state variable $\mathcal{S}_0$, which simply translates to a tenure of 0.

$$V_u(z,\chi) = \max_\theta \; b - \nu(\theta) + \beta(1-d)\begin{pmatrix} p(\theta,z)Pr(m(\chi))\, V_e^P(z,m(\chi),\mathcal{S}_0) \\ + \, p(\theta,z)(1-Pr(m(\chi)))\, V_e^{NP}(z,m(\chi),\mathcal{S}_0) \\ + \, (1-p(\theta,z))V_u(z,n(\chi)) \end{pmatrix} \tag{3.4}$$

In equation 3.4, $m(\chi)$ and $n(\chi)$ map the prior market belief χ into the post-search belief after success and failure, respectively (defined in Section 3.3). The function $Pr(\pi)$ maps the matched employer's belief into the promotion probability, allowing "promotion" upon hiring[10].

3.2 Employers and Beliefs

There are infinitely many homogeneous employers. When not matched with a worker, an employer posts vacancies at no cost, which are filled randomly. Therefore, the external market can be summarized by the set of functions $m(\chi), n(\chi) : [0,1] \to [0,1]$ and $\Gamma(\mathcal{S}) : [0,1] \times \mathbb{N}_0 \times \mathbb{N}_0 \to [0,1]$, which map initial beliefs and employment outcomes to new beliefs. I begin by describing matched employers and their belief process and then discuss the external belief process.

Matched Employers At the end of each period, with probability ρ, a matched employer reviews his employee's performance. A performance review at the end of period t means that the employer observes production for that period only, y_t. Upon this review, the employer updates his beliefs according to Bayes' rule. Equation 3.5 defines the update process, where the updated belief, π', is a function of the previous belief, π, and the worker's promotion status. $f(y|z,p)$ is the density function of production conditional on a worker's type and promotion status.

$$\pi' = \pi'(y|\pi,p) = \frac{\pi f(y|z=1,p)}{\pi f(y|z=1,p) + (1-\pi)f(y|z=0,p)} \tag{3.5}$$

Following the review, the employer makes two decisions: (i) whether to fire the worker (yielding value 0) and (ii) conditional on retention, whether to promote the worker if he is not already promoted. The firing and promotion decisions are given in equations 3.6 and 3.7, respectively.

$$F(\pi,p) = \begin{cases} 1 & \text{if } (1-\omega)\mathbb{E}[y(z,p) - k_p|\pi,p] < 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.6}$$

[10] A promotion of this form would not be observed in the data and will not be counted as a promotion in the simulation.

$$P^*(\pi, p) = \max_{j \in \{p,1\}} \left\{ \beta^P((1-\omega)\mathbb{E}[y(z,j) - k_j|\pi, j] - j\psi) + \epsilon_j \right\} \text{ where } \epsilon_j \sim \text{T1EV} \quad (3.7)$$

In equation 3.7, ψ represents a training cost incurred by the employer for promoting its employee. Note that the choice correspondence for the decision to promote a worker is the set $\{p, 1\}$, which reflects the assumption that the firm cannot choose to demote a worker. Because $\epsilon_j \sim$ T1EV, the employer's promotion rule is given by equation 3.8.

$$\begin{aligned} \Pr(\pi) &= \frac{e^{\beta^P U(\pi)}}{1 + e^{\beta^P U(\pi)}}, \\ \text{where} \quad U(\pi) &= (1-\omega)(\mathbb{E}[y(z,1) - k_1 \mid \pi] - \mathbb{E}[y(z,0) - k_0 \mid \pi]) - \psi. \end{aligned} \quad (3.8)$$

Figure 3.2 provides an example of the evolution of π over time for a given worker. Upon the formation of the match, π takes its initial value from the external labor market. If a performance review does not occur, π simply persists into the next period. If a review does occur, π evolves according to the equation for π', taking production in that period as the new source of information.

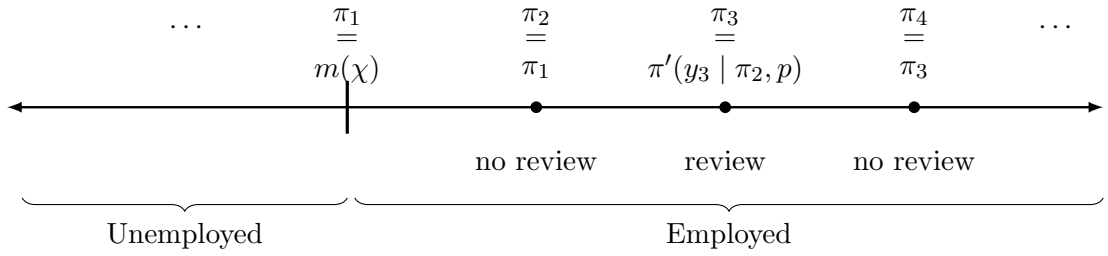


Figure 3.2: Evolution of π

Notes: This figure provides an example of how π, the current employer's belief about the worker, evolves over time. π' is defined in equation 3.5.

Non-Matched Employers The external labor market does not observe production directly, but it observes two outcomes that are informative about type: (i) job-finding success (or lack thereof) while unemployed and (ii) promotion histories across employment spells. Let $m(\chi)$, $n(\chi)$, and $\Gamma(\mathcal{S})$ map the market prior χ and observed histories into posterior beliefs. I begin by defining $m(\chi)$ and $n(\chi)$, which update beliefs following search success and failure.

Let $Pr(\pi)$ be the equilibrium promotion rule from equation 3.8 and let $h(z, \chi) \equiv p(\theta^*(z, \chi))$ be the equilibrium search policy from equation 3.4. Equations 3.9 and 3.10 define the corresponding belief updates: $m(\chi)$ is the posterior upon successful search, and $n(\chi)$ is the posterior upon unsuccessful search.

$$m(\chi) = \frac{\chi h(1,\chi)}{(1-\chi)h(0,\chi) + \chi h(1,\chi)} \tag{3.9}$$

$$n(\chi) = \frac{\chi(1-h(1,\chi))}{(1-\chi)(1-h(0,\chi)) + \chi(1-h(1,\chi))} \tag{3.10}$$

Updating χ from within-spell outcomes (e.g., a promotion) is not recursive because later events depend on intermediate performance signals. For example, consider a worker hired in period t who is promoted at the end of period $t+2$, and assume $\rho = 1$. Let χ denote the market belief at the start of the spell. Equation 3.11 gives the likelihood of this event.

$$\begin{aligned}\mathcal{L}(\chi, z) = \int_{-\infty}^{\infty} (1 - Pr(\pi'(y_1|\pi=\chi,0)))(1 - F(\pi'(y_1|\pi=\chi,0),0)) \int_{-\infty}^{\infty} \cdots \\ Pr(\pi'(y_2|\pi=\pi'(y_1|\pi=\chi,0),0))\phi(y_2|p=0,z)dy_2 \;\; \phi(y_1|p=0,z)dy_1\end{aligned} \tag{3.11}$$

Because the period-$t+1$ outcome is nested in the promotion probability at $t+2$, the external market cannot update recursively within the spell. Instead, it treats the entire spell as a single event and computes $\mathcal{L}$ when the worker enters unemployment[11].

Under the no-demotion assumption, the sufficient state variables for this update are χ, nu, and np. Here, nu is the number of periods spent non-promoted, and np is the number of periods spent promoted. Without this restriction, the external market would need the entire promotion-status history within the spell, a vector of size 2^K, where K is tenure in periods. To simplify notation, I separate $\Gamma(\chi, nu, np)$ into $\Gamma_P(\chi, nu, np)$ to cover cases in which $np > 0$ and $\Gamma_{NP}(\chi, nu)$ for cases in which $np = 0$. Under the no-demotion restriction, $\Gamma_{NP}(\chi, nu)$ and $\Gamma_P(\chi, nu, np)$ can be defined as in equations 3.12 and 3.13, respectively.

$$\Gamma_{NP}(\chi, nu) = \frac{\chi P_N^{(1)}(nu|z=1,\chi)}{\chi P_N^{(1)}(nu|z=1,\chi) + (1-\chi)P_N^{(1)}(nu|z=0,\chi)} \tag{3.12}$$

$$\Gamma_P(\chi, nu, np) = \frac{\chi P_P^{(1)}(nu, np|z=1,\chi)}{\chi P_P^{(1)}(nu, np|z=1,\chi) + (1-\chi)P_P^{(1)}(nu, np|z=0,\chi)} \tag{3.13}$$

$P_N^{(\tau)}(nu|\chi, z)$ is the probability that a type-z worker, matched with initial belief χ, is never promoted, works for nu periods, and then separates into unemployment. Similarly, $P_P^{(\tau)}(nu, np|\chi, z)$ is the probability that the worker spends nu periods non-promoted and

[11] This is as if the external market reads the worker's resume, which contains job titles and dates.

then np periods promoted before separation. Formal definitions are provided in Appendix A.

Figure 3.3 provides an example of how χ evolves over time for a given worker. During employment spells, χ remains unchanged (χ_0 and χ_3 in the provided example). Upon job separation, χ updates according to the function Γ. For example, if he was promoted, as in the first employment spell in Figure 3.3, it evolves according to Γ_P. While the worker is unemployed, χ evolves after each period. Following unsuccessful job searches, it evolves according to n, while successful searches result in an update according to m.

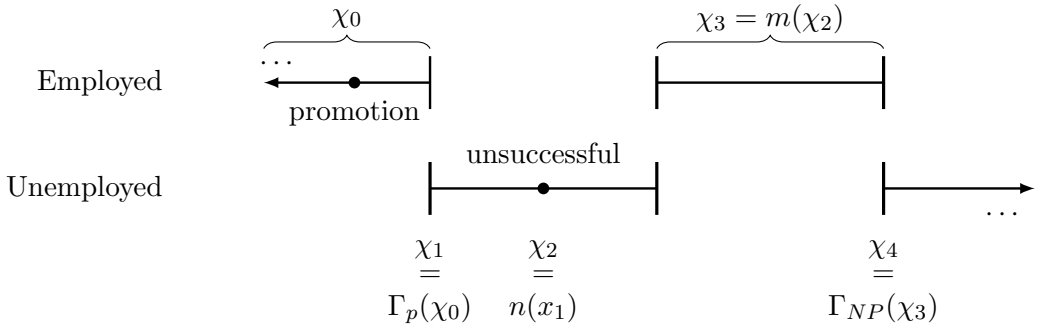


Figure 3.3: Evolution of χ

Notes: This figure provides an example of how χ, the "external market" belief of a worker, evolves over his career.

3.3 Equilibrium

Equilibrium is defined as a set of search policy functions $h(z, \chi)$, a promotion rule $\Pr(\pi)$, a firing rule $F(\pi, p)$, an employed and promoted value function $V_e^P(z, \pi, \chi, nu, np)$, an employed and non-promoted value function $V_e^{NP}(z, \pi, \chi, nu)$, an unemployed value function $V_u(z, \chi)$, and belief transition functions $n(\chi)$, $m(\chi)$, $\Gamma_P(nu, np, \chi)$, and $\Gamma_{NP}(nu, \chi)$ such that, given a production function $y(z, p)$ and a wage rule $w(\pi, p)$,

1. $h(z, \chi)$ solves the worker's search problem in equation 3.4 $\forall z, \chi$ and has the resulting value functions $V_u(z, \chi)$, $V_e^{NP}(z, \pi, \chi, np)$, and $V_e^P(z, \pi, \chi, nu, np)$;
2. $Pr(\pi)$ is the resulting choice-probability function from equation 3.7;
3. $F(\pi, p)$ is the solution to the firing rule in equation 3.6 $\forall \pi, p$; and
4. given $F(\pi, p)$, $Pr(\pi)$, and $h(z, \chi)$, $n(\chi)$, $m(\chi)$, $\Gamma_P(nu, np, \chi)$, and $\Gamma_{NP}(nu, \chi)$ are the Bayesian belief-updating functions for employers.

The model is solved via value-function iteration. First, use $y(z, p)$ and $w(\pi, p)$ to solve for $Pr(\pi)$ and $F(\pi, p)$. Using these, solve for $\Gamma_{NP}(nu, \chi)$ and $\Gamma_P(nu, np, \chi)$ as written in

Appendix A. Guess values for V_e^{NP}, V_e^P, V_u, and $h(\chi, z)$. Using the guess for $h(\chi, z)$, solve for $n(\chi)$ and $m(\chi)$ as written in equations 3.9 and 3.10. Appendix B provides a proof of existence.

4 Estimation and Model Fit

The model is estimated in two stages. First, I calibrate a set of parameters to match those provided in the prior literature or to match basic descriptive results. After this, I use the method of simulated moments to estimate parameters concerning the promotion rules, job search, and wages.

Tables 4.1 and 4.2 report the calibrated and estimated parameters, respectively. This section briefly motivates the calibration choices and then provides a more detailed discussion of how the estimated parameters are identified.

Calibration The unconditional distribution of worker types, Π, is estimated empirically using k-means clustering. For each worker in the NLSY, I construct four summary statistics: (i) average wage, (ii) total lifetime earnings, (iii) the number of unemployment spells, and (iv) the average duration of unemployment spells[12]. With $k = 2$, the algorithm partitions workers into two mutually exclusive groups by minimizing within-cluster Euclidean distance in this four-dimensional space. I interpret the resulting clusters as "low type" and "high type" workers, and set Π equal to the share of workers assigned to the high-type cluster.

The worker's share of expected profit, ω, is set to 0.5 to loosely match a wage-setting environment with equal bargaining power. The standard deviation of transitory shocks to log production, σ_y, is set to 0.5, in line with values commonly used in the literature.

The flow cost of maintaining a non-promoted position, k_0, is set to 0.355 so that firms are roughly indifferent between hiring a worker with an extremely low promotion posterior ($\pi \approx 0$) and remaining unmatched[13]. The parameters d and δ are chosen to match two moments in the NLSY: an average career length of 480 months (40 years) and an average of five unemployment spells over the life cycle. Home production (or unemployment benefits), b, is set to 40% of average wages. The probability of a performance review is set to $\rho = 1/12$, implying one review per year on average. Finally, γ is set to 1.5, and α_0 is normalized to 1.

The discount parameter, β, is low. A monthly β of 0.9 implies an annual discount factor of nearly 0.40. Because there are no accumulated variables (e.g., capital), the effect of β plays only a mechanical role in guaranteeing convergence. Increasing β to 0.99 (~ 0.89 annually) affects only the average duration of unemployment and can be completely offset by increasing ξ in the estimation procedure.

[12] For the small number of workers who never enter unemployment, I set the average duration to zero.

[13] Equivalently, k_0 is normalized to make the value of a low-type hire near the outside option of a vacancy.

Table 4.1: Calibrated Parameters

Parameter	Value	Description	Calibration Target
Π	0.38	% of high types	Estimated outside model
ω	0.5	wage piece rate	Mortensen and Pissarides (1994); Shimer (2005)
k_0	0.35	flow cost of un-promoted position (e.g. rent)	value of low-type worker to firm
σ_y	0.5	s.d. of output shocks	Abowd and Card (1989); Meghir and Pistaferri (2004)
γ	1.5	curvature of search cost	
ρ	1/12	Review probability	1 review a year
d	1/480	Prob. of death/retiring	avg. 40-year career
β	0.9	Discount factor	
δ	5/430	Exogenous separation probability	Average 5 separations per career
b	0.2	Unemployment benefits / home production	40% of average wages
α_0	1	Search-productivity of low-types	Normalized to 1

Notes: The table includes both calibrated parameters and parameters estimated empirically outside the model. Blank calibration targets indicate that the parameters are set arbitrarily, but the qualitative and quantitative results are robust to their effects.

Estimation After calibration, I estimate the remaining parameters using the method of simulated moments (MSM). In total, I estimate seven parameters: (i) the one-time cost of promotion (e.g., training), ψ; (ii) the loss in log production from failing to promote a high-type worker, c_0; (iii) the loss in log production from promoting a low-type worker, c_1; (iv) the weight the firm places on profits in the promotion decision, β^p; (v) the flow cost of maintaining a promoted position, k_1; (vi) the search productivity of high types, α_1; and (vii) the utility cost of search.

Let $m^{sim}(\theta)$ and m^{emp} denote $K \times 1$ vectors of simulated and empirical moments. The estimator solves

$$\hat{\theta} = \arg\min_{\theta} \big(m^{sim}(\theta) - m^{emp}\big)' W \big(m^{sim}(\theta) - m^{emp}\big), \tag{4.1}$$

where $W = \text{diag}\big(1/(m_k^{emp})^2\big)_{k=1}^K$. This weighting matrix normalizes each moment by its empirical value, so the criterion is equivalent to minimizing $\sum_{k=1}^K \big(m_k^{sim}(\theta)/m_k^{emp} - 1\big)^2$, placing all moments on a common scale regardless of their units.

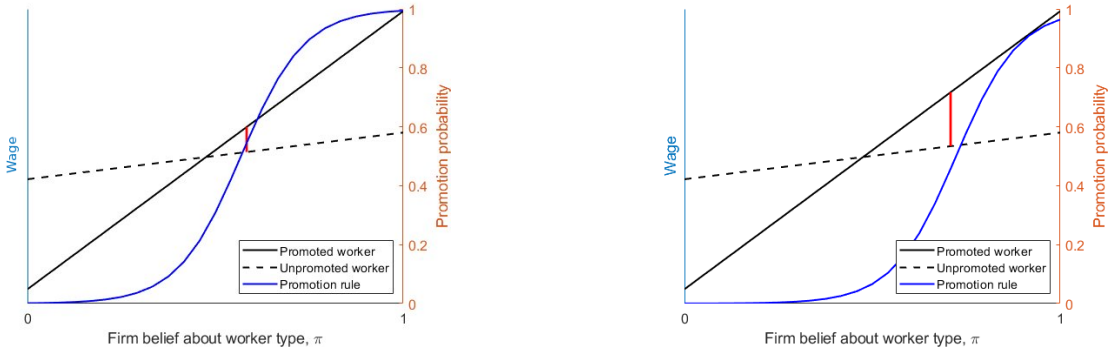
First, the set of parameters (ψ, c_0, c_1) is estimated to match Figure 2.3(b). The fixed cost ψ primarily governs the average wage jump at promotion. Production penalties directly govern the slope of the wage function before and after a promotion. Figure 4.1 provides a visual representation of the identification of ψ. Given a wage function, ψ pushes

Table 4.2: Estimated Parameters

Parameter	Estimate	Description	Identifying Moments
ψ	0.025	fixed cost of promotion	average promotion size
c_0	0.64	production penalty for "wrong" non-promotion	slope of wages pre-promotion
c_1	0.60	production penalty for "wrong" promotion	slope of wages post-promotion
β^p	30	Promotion-preference shocks	sd of promotions
k_1	0.87	flow-cost of promoted position	% promoted in first job
α_1	1.15	search productivity of high types	difference in average unemployment spell by type
ξ	0.42	utility cost of search	Average unemployment spell

Notes: Estimated parameters with their values, descriptions, and empirical targets. All values are estimated via the method of simulated moments.

the inflection point of the promotion function to the right. In both panels, the red line serves as a proxy for the median promotion size.



(a) "Small" Promotion Cost

(b) "Large" Promotion Cost

Figure 4.1: Identification of ψ

Notes: Panels A and B provide the wage functions for a worker with belief π and a given promotion status. The blue line represents the promotion function given the wage rule and the promotion cost, ψ. The average promotion roughly tracks where the promotion rule crosses the 50% mark. Panels A and B are identical except for the value of ψ, which is larger in Panel B.

The parameter β^P directly influences two separate moments that push in opposite directions. Thus, estimating β^P involves weighing the two moments and finding the ideal middle ground. Most obviously, β^P is identified from the dispersion of wage changes at promotion. As shown in Table 2.2, the 10th and 90th percentiles of wage changes at promotion are -12.2% and 51.76%, respectively. In the model, β^p governs how sharply the firm's promotion rule responds to changes in its belief about the worker. If $\beta^p \to \infty$, the promotion

decision becomes deterministic: the firm promotes the worker as soon as promotion yields positive profits and never otherwise. If $\beta^p = 0$, the promotion function is 0.5 $\forall\pi$, which generates excessive dispersion. Visually, decreasing β^P "flattens" the promotion function presented in Figure 4.1, which clearly increases dispersion. Unfortunately, decreasing β^P has adverse effects on the slope of wage loss with respect to unemployment duration. When $\beta^P = 0$, a previously promoted worker and a previously nonpromoted worker face identical expected wage functions: each is assigned to a promoted or nonpromoted position with probability 0.5 upon matching with a firm. Increasing β^P allows unemployment duration to affect promoted and nonpromoted workers heterogeneously.

Given k_0, the parameter k_1 governs the value of $\bar{\pi}$ as discussed in Figure 3.1. Therefore, as k_1 increases relative to k_0, it becomes rarer for a worker to be promoted by any given employer.

The search parameters α_1 and ξ are estimated to match unemployment duration by type. For this, I use the empirical type classification obtained from the aforementioned k-means procedure. After normalizing $\alpha_0 = 1$, α_1 governs the relative job-finding efficiency of high types and is therefore disciplined by the gap in average duration across types. The search-cost parameter ξ shifts equilibrium search effort for all workers and is therefore pinned down by the overall level of unemployment duration.

Model Fit The model matches the wage effects of promotions reasonably well. Figure 4.2 plots log wages in a 36-month window around each worker's first promotion. The solid lines reproduce the empirical event study from Figure 2.3(b), while the dashed lines show the model simulation. The model slightly overestimates the average effect of promotion (about 20% in the simulation versus 17% in the data), but it matches the average wage-growth rates before and after promotion reasonably well.

The parameter β^p targets the dispersion of promotion-induced wage changes in the NLSY. Specifically, β^p is set to match a standard deviation of 0.3496. However, due to the countervailing forces of the two moments that β^P targets, I match the gradient far better than the dispersion of promotions. The simulated standard deviation of promotions in the model is nearly 0.17—half of that in the data. While the model understates dispersion, Table 4.3 shows that this iteration is a notable improvement over the canonical promotion model from Gibbons and Waldman (1999), whose implied distribution is shown in the third row of Table 4.3. While further decreasing the value of β^P would improve the simulated 10th percentile of promotions, it would come at the expense of matching the wage-loss gradients in Figure 4.5.

The parameters α and ξ target the distribution of unemployment duration for both high and low types. In the data, average unemployment duration is 8.25 months for high types and 11.41 months for low types. In the simulated data, the corresponding averages are 7.27 and 11.14 months.

Figure 4.3 presents the model fit around unemployment-to-employment (U2E) transitions. The targeted moments are the discontinuous drops at time 0 (i.e., at re-employment),

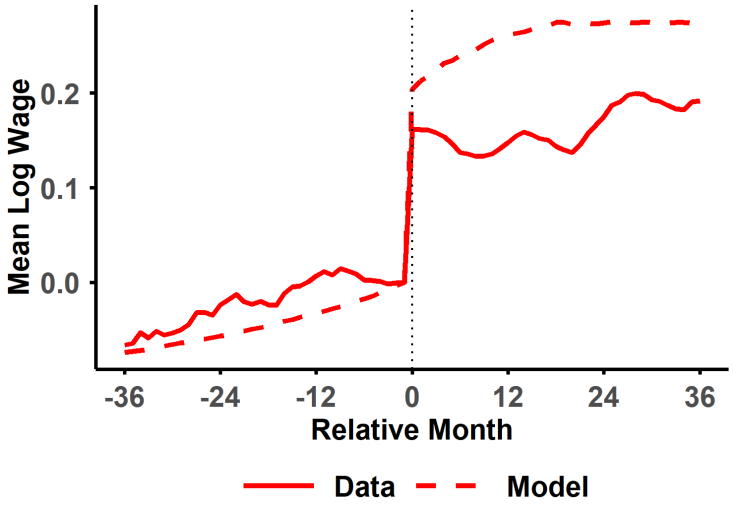


Figure 4.2: Wage Effect of Promotions: Model and Data

Notes: The figure represents log wages around all first promotions in the NLSY and in data simulated from the model. The dashed line represents the model, and the solid line represents the data, identical to those shown in Figure 2.3(b).

Table 4.3: Distribution of Promotions: Model and Data

	p10	p20	p50	p80	p90
Data	-12.2	1.6	14.7	34.8	51.8
Model	2.69	9.73	21.91	32.57	37.34
Model$_{\beta^p \approx \infty}$	13.9	17.5	25.5	34.1	38.4

Notes: Values represent the 10th, 20th, 50th, 80th, and 90th percentiles of promotion-induced wage gains.

which the model matches closely. As an untargeted moment, the model does reasonably well in matching the path of wages leading into the unemployment spell, but it cannot quite match the path of wages after re-entry into employment.

The parameter k_1 is estimated to match the share of workers who are promoted prior to entering unemployment—that is, the relative size of the promoted group in the U2E event studies. Currently, the model understates this promotion rate: the empirical share is about 34%, whereas the simulated share is just over 19%. However, as an untargeted moment, the model does quite well in matching the life-cycle of promotion rates. Figure 4.4 plots both the empirical (panel a) and the simulated (panel b) promotion rate by age. The model is capable of replicating the steep decline of promotions early in the career, though it does move to the bottom ($\sim 3\%$) earlier than the data.

The information mechanism also allows the model to naturally match the empirical interaction between promotions and unemployment duration. Figure 4.5 plots the simulated analog of Figure 2.10. While the model underpredicts the depreciation rate for both

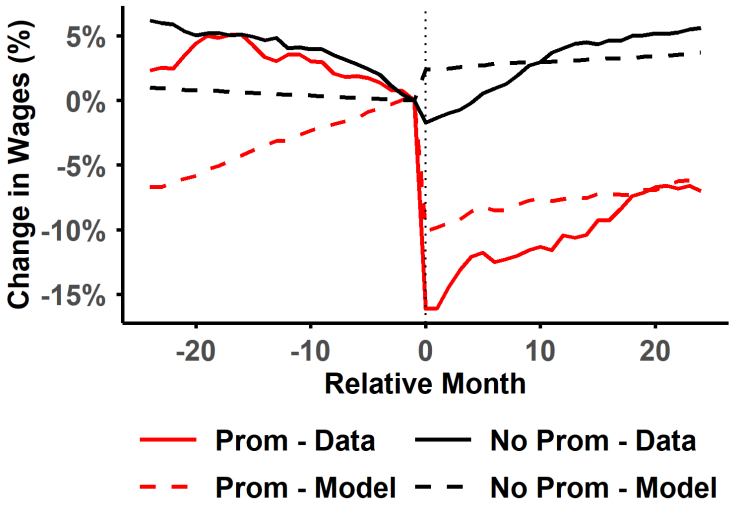
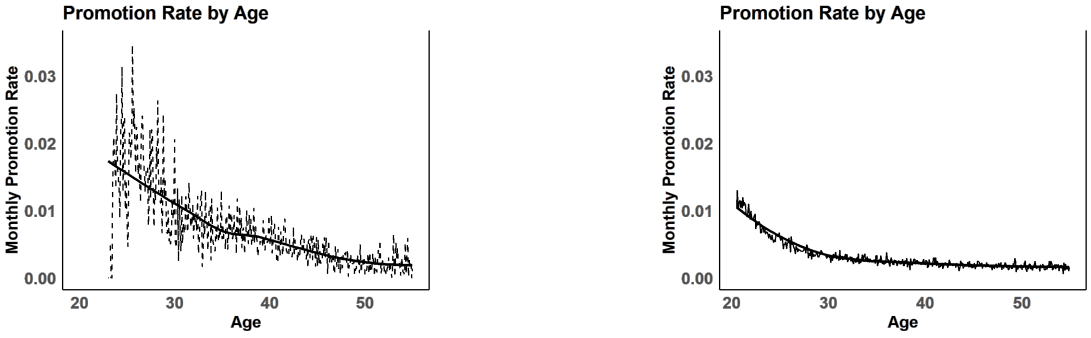


Figure 4.3: Wages Around U2E Transitions: Model and Data

Notes: The figure represents log wages around each unemployment spell in both the model and the data. Dotted lines represent data simulated from the model, while solid lines represent data from the NLSY.



(a) Data

(b) Model

Figure 4.4: Promotion Rate by Age: Model and Data

Notes: The panels provide the monthly promotion rate from the data (a) and the model (b). While the model underpredicts the overall frequency of promotions, it can match the shape without specifically targeting it.

groups, it captures the key qualitative fact that unemployment duration harms previously promoted workers much more than it harms non-promoted workers.

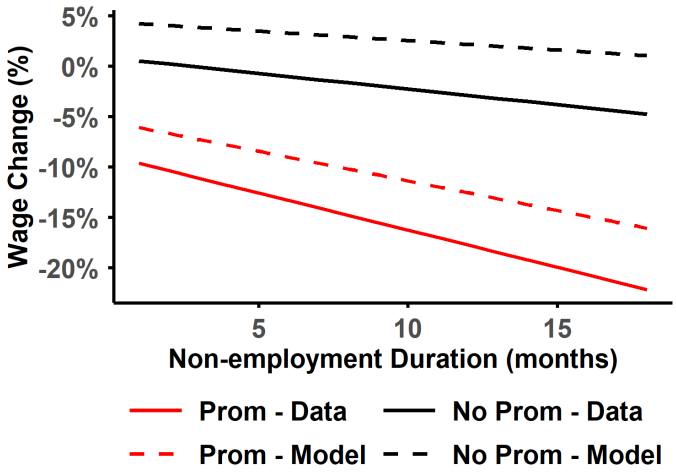


Figure 4.5: Wage Loss and Duration: Model and Data

Notes: The figure plots the relationship between unemployment duration and promotion status in both the model (solid) and the data (dashed).

5 Labor Market Sorting

Labor market sorting in this model occurs through two mechanisms. While employing a worker in a low-type position, the firm may choose to promote him by moving him up into a high-type position. The firm makes this decision only when it has gathered sufficient information suggesting that the worker is high-type. On the other hand, downward positional mobility occurs via unemployment. When a firm matches with an unemployed worker who previously worked in a high-type position, it chooses whether to "re-assign" him back down into a low-type position[14]. Firms use unemployment duration as a source of information to guide this decision because high-type workers match with greater probability than their low-type counterparts.

In this section, I show how firms' ability to properly "re-assign" workers upon the formation of a match is affected by two separate unemployment policies, both of which work by increasing or decreasing the incentives for workers of different types to exert effort in job search. Policies that incentivize a convergence of search effort across worker types reduce the potency of unemployment duration as a signal. For example, suppose the government enacted a policy such that $p(\theta^*(\chi, z = 1)) = p(\theta^*(\chi, z = 0))$. In such a

[14]Firms do have the ability to "re-assign" in the opposite direction, but because both unemployment and non-promotions are negative signals, this almost never happens within the model.

case, the belief functions in equations 3.9 and 3.10 revert to $m(\chi) = n(\chi) = \chi$, and the rate at which firms make the wrong re-assignment decision increases. Similarly, policies that encourage search efforts resulting in $p(\theta^*(\chi, z = 1)) >> p(\theta^*(\chi, z = 0))$ increase the potency of information, as $|m(\chi) - \chi|$ and $|n(\chi) - \chi|$ both increase.

There are two policies I explore to target this mechanism. The first, which I call the FC (fixed-cost) policy, allows the government to condition unemployment benefits on a fixed initial payment from the worker when he separates from his employer[15]. Workers who choose to pay the upfront cost are eligible to receive benefits for the duration of their unemployment spell, while those who do not receive nothing. Intuitively, workers who receive benefits will exert less effort and match with lower probability than those who do not. Because low-type workers expect longer spells, they are more willing to pay the upfront cost than their high-type counterparts. Thus, with the right fixed cost, many high-type workers will opt out of receiving benefits, while low-type workers will not. The second policy adjusts the UI function in the following way: in the baseline model, all unemployed workers receive benefits b each period, where b is calibrated to equal 40% of the average wage, such that $b = 0.4\bar{w}$. Under the new policy regime, the benefits paid to worker i, b_i, are a function of the worker's most recent wage, w_T, such that $b_i = 0.4w_T$. While firms do not observe a worker's previous wages, they know that high-type workers are more likely to have had a high wage before separation. Because of this, high-type workers will, on average, receive greater benefits than their low-type counterparts. This results in decreased search effort among high types and increased effort among low types.

I benchmark the performance of the policies against a world of symmetric information. Recall that, in the baseline model, "external" firms do not observe worker performance (output) directly. Instead, they observe the position-assignment decisions of a worker's previous employers and use them as coarse signals of his performance in his prior matches. With symmetric information, I allow output to be a public signal to all firms, which results in far quicker learning and more desirable sorting outcomes. As such, symmetric information serves as an upper bound on firm performance in sorting.

Panel (a) of Table 5.1 reports the re-assignment performance of firms across these different policy regimes, and the symmetric information benchmark. Error-H is defined as the percent of high-type workers that are re-assigned incorrectly, while Error-L is the percent of low-type workers that are re-assigned incorrectly. Thus, in the baseline model, 32.49% of high-type workers in high-type positions are "re-assigned" into a low-type position after unemployment, while 18.73% of low-type workers in high-type positions are incorrectly retained in a high-type position upon the formation of a new match. When worker output is public, both error rates drop substantially, to 6.98% and 11.93%, respectively.

The rate at which firms err in their re-assignment decisions is sensitive to the policy environment. Under the FC regime, error-H is reduced to 23% at the expense of an increase in error-L to just over 33%. Thus, upon re-employment, firms are more likely to preserve

[15] I take no stand on the source of this payment. It can be dollars, utility, etc.

a worker's position as he transitions through an unemployment spell. The same trade-off is present under the proportional UI regime, where error-H increases to nearly 34% and error-L decreases to 17.6%.

Panel (b) presents average output and wages in each regime relative to those in the baseline. That is, each column represents the relative change from the baseline value of the same column. Columns 1, 4, and 7 show the average effects across all workers, while the rest are conditional on worker type. Turning attention to columns 1–3, note that the effect on overall output is substantial: when firms are symmetrically informed, aggregate output increases by 4.26%, primarily driven by a drastic increase in output from high-type workers, while total output from low-type workers decreases by 1.33%. This decrease is due to an increase in the overall unemployment rate of low-type workers, which is clear upon investigating columns 4–6. Columns 4–6 document average output for workers conditional on being employed. Column 6 of row 2 shows that, conditional on being employed, low-type workers experience an increase in output, reflecting the fact that they are more accurately sorted into low-type positions. Wages increase by 2.7% on average, again driven entirely by high-type workers. Low-type workers experience a decrease in average wages because fewer are able to successfully "fool" firms into assigning them to high-type positions.

Within the FC regime, we see that the improvement in sorting documented in panel (a) is reflected in output and wages. Aggregate output increases by 2.5%, and average wages, conditional on being employed, increase by over 4%. Interestingly, columns 8 and 9 show that wages increase for *both* worker types. As seen in panel (a), firms under the FC regime are less willing to remove a high-type assignment from a given worker. As a result, both high-type and low-type workers more frequently benefit from the wage premium offered by the position. For low-type workers, this comes at the expense of the firm. Column 6 of row 3 notes that employed low-type workers generate 0.9% *less* output relative to the baseline despite earning higher wages.

Finally, row 4 documents the output and wage effects of the proportional UI regime. Aggregate output decreases by over a quarter of a percent due to worsened sorting of high-type workers, as employed high-type workers produce over half a percent less output.

Each environment (Sym, FC, Prop) affects aggregate employment through three channels: the unemployment rate, the composition of the workforce, and sorting. I decompose each regime's effect on aggregate output using the Shapley-value decomposition. Specifically, steady-state aggregate output can be written as in equation 5.1, where Y takes as arguments the steady-state unemployment rate of regime r, the percentage of all employed workers who are high-type, R, the percentage of employed high-type workers in high-type positions, α, and the percentage of low-type workers in high-type positions, β, and returns aggregate output.

$$Y(U_r, R_r, \alpha_r, \beta_r) = (1-U)\begin{pmatrix} R\big(\alpha_r \mathbb{E}[y|z=1,p=1] + (1-\alpha_r)\mathbb{E}[y|z=1,p=0]\big) + \\ (1-R)\big(\beta_r \mathbb{E}[y|z=0,p=1] + (1-\beta_r)\mathbb{E}[y|z=0,p=0]\big) \end{pmatrix} \quad (5.1)$$

Table 5.1: Performance by Policy Regime

Panel A: Error in Firm Re-assignment

Policy	Error-H	Error-L
Base	32.49	18.73
Sym	6.98	11.93
FC	22.94	33.14
Prop	33.97	17.57

Panel B: Output and Wage Moments

	Av. Output			Av. Output | Emp.			Av. Wages		
Policy	All	High	Low	All	High	Low	All	High	Low
Base	—	—	—	—	—	—	—	—	—
Sym	4.26%	8.20%	−1.33%	4.86%	7.24%	0.18%	2.70%	8.70%	−4.12%
FC	2.50%	4.93%	−1.02%	2.28%	4.16%	−0.89%	4.32%	4.62%	3.89%
Prop	−0.27%	−1.05%	1.02%	−0.54%	−0.32%	0.00%	−0.36%	−0.41%	0.23%

Notes: Panel A provides firms error rate in re-sorting decisions. Error-H is the rate at which firms incorrectly assign a high-type worker into a low-type position at the formation of a new match, while Error-L is the rate at which firms incorrectly keep a low-type worker in a high-type position at the formation of a new match. Panel-B presents the effect on output and wages, relative to the baseline model. In both panels, policy "Sym" is the world where performance is public to all firms, "FC" is the policy where workers pay a fixed cost to be eligible for unemployment benefits, and "Prop" is the policy where UI benefits are dependent on the worker's history of wages.

These three values summarize the effects of unemployment, labor-force composition, and sorting, respectively. Thus, the Shapley-value decomposition estimates the marginal effect of a given mechanism by averaging the change in $\log Y$ over all permutations of its inputs. For example, consider two regimes, $r = b$ and $r = FC$. The effect of unemployment when going from b to FC can be written as follows:

$$ME_U(b, FC) = \begin{pmatrix} 2(\log Y(U_{FC}, R_b, \alpha_b, \beta_b) - \log Y(U_b, R_b, \alpha_b, \beta_b)) + \\ \log Y(U_{FC}, R_{FC}, \alpha_b, \beta_b) - \log Y(U_b, R_{FC}, \alpha_b, \beta_b) + \\ \log Y(U_{FC}, R_b, \alpha_{FC}, \beta_{FC}) - \log Y(U_b, R_b, \alpha_{FC}, \beta_{FC}) + \\ 2(\log Y(U_{FC}, R_{FC}, \alpha_{FC}, \beta_{FC}) - \log Y(U_b, R_{FC}, \alpha_{FC}, \beta_{FC})) \end{pmatrix} /6$$

The results of this decomposition are presented in Table 5.2. In the first row, cumulative, I provide the relative change in aggregate output represented by $\log Y(U', R', \alpha', \beta') - \log Y(U, R, \alpha, \beta)$. These are approximations of the values provided in column 1 of Table 5.1. Each column in Table 5.2 decomposes the cumulative effect into its three components, which are shown in the rows beneath it. It is clear that, in all three regimes, sorting dominates the effect on output. Holding fixed the unemployment rate and the labor-force composition, changing the sorting performance from the baseline model to the symmetric regime

results in a 4.2% increase in aggregate production, nearly 100% of the cumulative effect. Similarly, the sorting mechanism is responsible for over 90% of the rise in output due to the FC regime and 90% of the output effect due to the proportional UI regime.

Table 5.2: Shapley-value Decomposition of Aggregate Output

	Sym	FC	Prop
Cumulative	4.16%	2.20%	−0.22%
Unemployment	−0.20%	0.10%	0.12%
Composition	0.20%	0.10%	−0.14%
Sorting	4.20%	2.00%	−0.20%

Notes: Table 5.2 decomposes the effect of each regime into its components. Row 1 "Cumulative" provides the relative change in aggregate output to the baseline model, while the rows below it provide the marginal effect of the change in the unemployment rate, the labor-force composition, and the sorting of each regime.

6 Conclusion

This paper studies how promotion within the firm and positional assignment after unemployment interact when information about worker ability is asymmetric. A promotion not only increases a worker's compensation, but it also serves as a signal of worker ability to the rest of the labor market. That signal remains valuable when the worker moves directly to another employer, yet become vulnerable to reinterpretation after job loss. Unemployment is therefore not merely time spent outside production, but is an informative signal through which prospective employers acquire information and may potentially reverse a worker's previous assignment.

The empirical analysis provides evidence for this interaction using the NLSY79. Promotions are associated with large immediate wage gains, but their long-run value depends strongly on subsequent attachment to employment. Promoted workers who remain employed or move directly between employers largely retain these gains. In contrast, previously promoted workers experience sizable wage losses after unemployment, while losses are much smaller for workers who were not promoted before separation. Moreover, the wage penalty for promoted workers increases sharply with unemployment duration. These findings suggest that the cost of unemployment depends on the worker's prior position and career history, beyond the effects of lost tenure, occupational change, or time out of work.

I interpret these patterns through an estimated model in which firms learn privately about worker type, promote workers using that information, and rely on publicly observed employment histories when hiring. Outside firms cannot perfectly distinguish an exogenous separation from a firing, and continued unsuccessful search is an additional negative

signal because high-type workers find jobs more quickly. Previously promoted workers consequently risk reassignment from high-type to low-type positions following unemployment. Quantitatively, asymmetric information generates substantial assignment errors and lowers production relative to an environment in which worker output is publicly observed.

In this environment, unemployment insurance affects this re-assignment process by changing the informativeness of unemployment duration. Benefits proportional to past wages reduce the search incentives of workers who are more likely to be high-type, thus compressing any differences in job-finding rates across types, and worsen the assignment of high-type workers. In the estimated model, this policy reduces aggregate output by approximately 0.27% relative to a flat benefit profile. Conversely, a fixed-cost policy that induces more high-type workers to forgo benefits, while preserving insurance for workers expecting longer spells, strengthens the relationship between type and duration and raises aggregate output by approximately 2.5%. The decomposition shows that these output effects operate primarily through sorting rather than changes in unemployment or workforce composition.

These results do not imply that unemployment policy should be evaluated only through sorting. Insurance, consumption smoothing, distributional objectives, and implementation costs remain essential considerations. Instead, the analysis identifies an additional policy margin: when firms learn from employment histories, policies that change search behavior also change the information available to employers. Future work could incorporate richer worker heterogeneity and study policies that communicate separation reasons or certify prior performance. More broadly, the paper shows that internal career decisions and external labor-market transitions cannot be studied independently. Promotions shape how unemployment is interpreted, while unemployment determines whether the allocation achieved within one firm survives into the next.

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A Likelihood of Spells

$$P_N^{(\tau)}(nu \mid z, \chi) = \begin{cases} \int \Big[\delta + (1-\delta)\rho F(\pi'(y_\tau \mid \chi, p=0), 0)\Big]\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \tau = nu, \\ (1-\delta) \int (1-\rho) P_N^{(\tau+1)}(nu \mid z, \chi) + & \tau \in \{1, \ldots, nu-1\} \setminus \{nu\}. \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) \times & \\ \qquad \big(1 - \Pr(\pi'(y_\tau \mid \chi, p=0))\big) \times & \\ \qquad P_N^{(\tau+1)}(nu \mid z, \pi'(y_\tau \mid \chi, p=0))\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \end{cases} \tag{A.1}$$

$$P_P^{(\tau)}(nu, np \mid z, \chi) = \begin{cases} \int \Big[\delta + (1-\delta)\rho F(\pi'(y_\tau \mid \chi, p=1), 1)\Big] & \tau = nu + np, \\ \quad \times\, \phi(y_\tau \mid p=1, z)\, dy_\tau & \\ (1-\delta) \int (1-\rho) P_P^{(\tau+1)}(nu, np | z, \chi) + & \tau \in \begin{matrix}\{nu+1, ..., nu+np-1\}/ \\ \{nu+np\}\end{matrix}, \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=1), 1)\big) \times & \\ \qquad P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=1))\, \phi(y_\tau \mid p=1, z)\, dy_\tau & \\ (1-\delta)\rho \int \big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) & \tau = nu, \\ \qquad \times \Pr(\pi'(y_\tau \mid \chi, p=0)) & \\ \qquad \times P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=0)) & \\ \qquad \times \phi(y_\tau \mid p=0, z)\, dy_\tau & \\ (1-\delta) \int (1-\rho) P_P^{(\tau+1)}(nu, np | z, \chi) + & \tau \in \{1, ..., nu-1\}/\{nu\}. \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) \times & \\ \qquad \big(1 - \Pr(\pi'(y_\tau \mid \chi, p=0))\big) \times & \\ \qquad P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=0))\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \end{cases} \tag{A.2}$$

Equations A.1 and A.2 calculate the probability that an employed worker of type z works in a non-promoted position for nu periods, works in a promoted position (if applicable) for np periods, and then loses his job, given that he started with an initial belief of χ.

As an example, suppose a worker enters the unemployment market after accruing $nu = 2$ and $np = 2$. This worker was promoted at the end of his second period and lost his job at the end of his fourth. Let χ be the belief with which he entered this unemployment spell. The value $P_P^{(1)}(2, 2|z, \chi)$ represents the likelihood that (nu, np) occurred for a worker of type z. Expanding $P_P^{(1)}(2, 2|z, \chi)$, as defined in equation A.2, would result in a four-level nested integral in which each step represents the event that occurred in the worker's k^{th} period of tenure, integrated over all paths that may have led there and weighted by their probabilities.

The recursive nature of this problem allows a programmer to build a mapping of (nu, np, z, χ) into new beliefs via backward induction.

B Existence of a Fixed Point Solution

This appendix establishes the existence of a fixed point $(V_u, V_e^P, V_e^{NP}, m^*, n^*)$ of the full system. Write the value function compactly as

$$V(z, \pi, \chi, nu, np, j) \;=\; \begin{cases} V_u(z, \chi), & j = 1, \\ V_e^{NP}(z, \pi, \chi, nu), & j = 2, \\ V_e^{P}(z, \pi, \chi, nu, np), & j = 3, \end{cases}$$

and let $S = Z \times [0,1] \times [0,1] \times \mathbb{N}_0 \times \mathbb{N} \times \mathcal{J}$ be the state space, where $Z = \{0,1\}$ and $\mathcal{J} = \{1,2,3\}$ carry the discrete topology and $[0,1]$ the standard topology. Although Z, $\mathcal{J}$, and $[0,1]^2$ are compact, $\mathbb{N}_0$ and $\mathbb{N}$ are not compact under the discrete topology, so S is not compact. We therefore work with $C_b(S)$—the Banach space of *bounded* continuous functions on S with sup-norm $\|f\| = \sup_{s \in S} |f(s)|$. Since the discrete topology makes every real-valued function continuous in the nu and np components, continuity in $C_b(S)$ reduces to continuity in (π, χ) for each fixed (z, nu, np, j). Bounded rewards and $\tilde{\beta} < 1$ ensure T maps $C_b(S)$ into $C_b(S)$.

The argument proceeds via three lemmas. Lemma B.5 establishes that the Bellman operator is a contraction for fixed belief transition functions, delivering a unique value function $V^*(m,n)$. Lemma B.6 shows that $V^*(m,n)$ and the induced optimal policy $\hat{\theta}^*(m,n)$ vary continuously with (m,n). Lemma B.9 constructs a compact convex set K_{L^*} that is mapped into itself by the belief-updating operator Ψ. Theorem B.11 then applies Schauder's fixed point theorem.

We maintain the following assumptions.

Assumption B.1. The functions $p(\theta, z)$, $\nu(\theta)$, $w(\pi, p)$, and $\Pr(\pi)$ are continuous in their arguments. Moreover, $\nu(\theta)$ is strictly convex and twice continuously differentiable, with $\nu(\theta) \to \infty$ as $\theta \to \infty$.

Assumption B.2. The matching probability satisfies

$$p(\theta, z) \;=\; \underline{p} + (1 - \underline{p})\big(1 - e^{-\alpha(z)\theta}\big) \;\in\; (\underline{p},\, 1),$$

where $\underline{p} > 0$ and $\alpha(z) \geq 1$ for all $z \in Z$. Set $\alpha^{\max} = \max_{z \in Z} \alpha(z)$ and $L_p^{\max} = (1 - \underline{p})\alpha^{\max}$; the latter serves as a uniform Lipschitz constant for p in θ. Since $\nu(\theta) \to \infty$ bounds optimal effort by some $\Theta^{\max} < \infty$, set $\bar{p} = \sup_{\theta \in [0, \Theta^{\max}], z} p(\theta, z) < 1$ and $\varepsilon_p = \min(\underline{p}, 1 - \bar{p}) > 0$.

Remark B.3. Assumption B.2 modifies the matching technology of the implemented model, which uses $p(\theta, z) = 1 - e^{-\alpha_z \theta}$ and satisfies $\inf_{\theta \geq 0} p = 0$. The positive floor $\underline{p} > 0$ is

introduced here because the Bayesian updating denominators in (B.1) require a strictly positive lower bound. The existence theorem therefore applies to this modified technology and not directly to the estimated model.

Assumption B.4. The optimal search effort satisfies $\hat{\theta}^*(z, \chi; m, n) > 0$ for all (z, χ, m, n).

Assumption B.4 is imposed independently. At $\theta = 0$, the marginal benefit of search equals $(1 - \underline{p})\alpha(z)\tilde{\beta}\,\Delta V(z, \chi)$, where $\Delta V(z, \chi)$ is the value surplus from a successful match. Positive optimal effort therefore requires $\Delta V(z, \chi) > 0$ for all (z, χ), which is not implied by Assumption B.2 alone and must be verified at the solution.

Lemma B.5 (Existence and uniqueness of V^*)**.** *For any fixed $(m, n) \in C([0, 1], [0, 1])^2$, the Bellman operator $T : C_b(S) \to C_b(S)$ is a contraction with modulus $\tilde{\beta} \equiv \beta(1 - d) \in (0, 1)$. Consequently, there exists a unique $V^*(m, n) \in C_b(S)$ satisfying $TV^*(m, n) = V^*(m, n)$.*

Proof. We verify Blackwell's sufficient conditions.

Monotonicity. Fix $v, t \in C(S)$ with $v \leq t$ pointwise, and let v_j, t_j denote their restrictions to employment state j. For $j = 1$, let $\hat{\theta}$ attain the maximum in (Tv):

$$\begin{aligned}
(Tv)(z, \pi, \chi, nu, np, 1) &= b - \nu(\hat{\theta}) + \tilde{\beta}\big[p(\hat{\theta}, z)\Pr(m(\chi))\, v_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad + p(\hat{\theta}, z)(1 - \Pr(m(\chi)))\, v_2(z, m(\chi), m(\chi), 0) \\
&\qquad + (1 - p(\hat{\theta}, z))\, v_1(z, n(\chi))\big] \\
&\leq b - \nu(\hat{\theta}) + \tilde{\beta}\big[p(\hat{\theta}, z)\Pr(m(\chi))\, t_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad + p(\hat{\theta}, z)(1 - \Pr(m(\chi)))\, t_2(z, m(\chi), m(\chi), 0) \\
&\qquad + (1 - p(\hat{\theta}, z))\, t_1(z, n(\chi))\big] \\
&\leq \max_{\theta}\Big\{ b - \nu(\theta) + \tilde{\beta}\big[p(\theta, z)\Pr(m(\chi))\, t_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad\qquad + p(\theta, z)(1 - \Pr(m(\chi)))\, t_2(z, m(\chi), m(\chi), 0) \\
&\qquad\qquad + (1 - p(\theta, z))\, t_1(z, n(\chi))\big] \Big\} \\
&= (Tt)(z, \pi, \chi, nu, np, 1).
\end{aligned}$$

The first inequality uses $p(\cdot), \Pr(\cdot) \geq 0$ and $v \leq t$ pointwise; the second uses feasibility of $\hat{\theta}$ for t. The argument is identical for $j = 2, 3$, establishing $Tv \leq Tt$.

Discounting. For any $v \in C(S)$ and constant $a \geq 0$, write $\mathbf{a}$ for the function identically equal to a:

$$\begin{aligned}
T(v + \mathbf{a})(s) &= \max_{\theta}\big[b - \nu(\theta) + \tilde{\beta}\,\mathbb{E}[(v + \mathbf{a})(s') \mid \theta, s]\big] \\
&= \max_{\theta}\big[b - \nu(\theta) + \tilde{\beta}\,\mathbb{E}[v(s') \mid \theta, s] + \tilde{\beta}a\big] \;=\; (Tv)(s) + \tilde{\beta}a.
\end{aligned}$$

The same calculation holds for $j = 2, 3$. Since $\tilde{\beta} \in (0, 1)$, the discounting condition is satisfied with modulus $\tilde{\beta}$.

By Blackwell's theorem, T is a contraction on $(C_b(S), \|\cdot\|)$ with modulus $\tilde{\beta}$. T maps $C_b(S)$ into itself: bounded rewards and $\tilde{\beta} < 1$ imply $\|Tv\| \leq \|b \vee w\|/(1 - \tilde{\beta})$ for any $v \in C_b(S)$; continuity in (π, χ) follows from Berge's maximum theorem. The Banach fixed point theorem yields the unique fixed point $V^*(m, n)$. □

Lemma B.6 (Continuity of V^* and $\hat{\theta}^*$)**.** *Let $(m_k, n_k) \to (m, n)$ uniformly in $C([0, 1], [0, 1])^2$. Then $\|V^*(m_k, n_k) - V^*(m, n)\| \to 0$, and $\hat{\theta}^*(\cdot\,; m_k, n_k) \to \hat{\theta}^*(\cdot\,; m, n)$ uniformly.*

Proof. Continuity of V^*. Write $V_k^* = V^*(m_k, n_k)$ and $V^* = V^*(m, n)$. Since both are fixed points, the triangle inequality gives

$$\begin{aligned}\|V_k^* - V^*\| &= \|T(V_k^*; m_k, n_k) - T(V^*; m, n)\| \\ &\leq \|T(V_k^*; m_k, n_k) - T(V^*; m_k, n_k)\| + \|T(V^*; m_k, n_k) - T(V^*; m, n)\| \\ &\leq \tilde{\beta}\|V_k^* - V^*\| + \|T(V^*; m_k, n_k) - T(V^*; m, n)\|,\end{aligned}$$

where the last line applies the contraction bound to the first term. Rearranging:

$$(1 - \tilde{\beta})\|V_k^* - V^*\| \leq \|T(V^*; m_k, n_k) - T(V^*; m, n)\|.$$

The right side vanishes. For employment states $j = 2, 3$, the Bellman operator involves only matched-employer dynamics and the pre-computed belief-update functions Γ; neither depends on the external belief-updating functions (m, n), so $T(V^*; m_k, n_k) = T(V^*; m, n)$ on those states. For the unemployment state $j = 1$, the operator depends on (m, n) only through $m(\chi)$ and $n(\chi)$ as arguments of V^*: the continuation values are $V^*(z, m(\chi), m(\chi), 0, 0, 3)$, $V^*(z, m(\chi), m(\chi), 0, 2)$, and $V^*(z, n(\chi), 1)$. Because $V^* \in C_b(S)$ and for each fixed discrete state the map $(\pi, \chi) \mapsto V^*(\cdot)$ is continuous on the compact domain $[0, 1]^2$ (hence uniformly continuous), uniform convergence of $m_k \to m$ and $n_k \to n$ yields $\|T(V^*; m_k, n_k) - T(V^*; m, n)\| \to 0$. Therefore $\|V_k^* - V^*\| \to 0$.

Continuity of $\hat{\theta}^*$. The search problem for $j = 1$ is $\hat{\theta}^*(\chi, z; m, n) = \arg\max_{\theta \in [0, \Theta^{\max}]} W(\theta, \chi, z, m, n)$, where the truncation to $\Theta^{\max} < \infty$ is without loss because $\nu(\theta) \to \infty$ and V^* is bounded. The objective W is jointly continuous in (θ, χ, m, n) by continuity of p, ν, Pr, and V^*. The constraint correspondence $\Phi(\cdot) = [0, \Theta^{\max}]$ is constant, hence trivially continuous and compact-valued. By Berge's maximum theorem, $\hat{\theta}^*$ is upper hemicontinuous in (m, n). Strict concavity of W in θ—which follows from the structure of p in Assumption B.2 and strict convexity of ν in Assumption B.1—implies the maximizer is unique at every parameter value. A single-valued upper hemicontinuous correspondence is a continuous function. □

Define the belief-updating operator $\Psi : C([0, 1], [0, 1])^2 \to C([0, 1], [0, 1])^2$ by $\Psi(m, n) = (m', n')$, where

$$m'(\chi) = \frac{\chi\, a(\chi)}{\chi\, a(\chi) + (1 - \chi)\, b(\chi)}, \qquad n'(\chi) = \frac{\chi(1 - a(\chi))}{\chi(1 - a(\chi)) + (1 - \chi)(1 - b(\chi))}, \tag{B.1}$$

$a(\chi) = p(\hat{\theta}^*(\chi, z{=}1; m, n), z{=}1)$, and $b(\chi) = p(\hat{\theta}^*(\chi, z{=}0; m, n), z{=}0)$. A fixed point of Ψ is a pair of belief transition functions consistent with optimal search behavior via Bayes' rule. For $L > 0$, define

$$K_L = \big\{(m,n) \in C([0,1],[0,1])^2 : |m(\chi){-}m(\chi')| \le L|\chi{-}\chi'|,\ |n(\chi){-}n(\chi')| \le L|\chi{-}\chi'|\ \forall\, \chi, \chi'\big\}.$$

The Lipschitz condition is preserved under convex combinations, and K_L is uniformly bounded and equicontinuous, so K_L is a compact convex subset of $C([0,1])^2$ by the Arzelà–Ascoli theorem.

Assumption B.7. For all $(m,n) \in K_L$, the fixed-point value function $V^*(m,n)$ is continuously differentiable in (π,χ) with derivatives bounded uniformly over K_L: $\|\partial V_j^*/\partial m\|_\infty, \|\partial V_j^*/\partial n\|_\infty \le D$ for some $D < \infty$ and $j \in \{1,2,3\}$. Moreover, the second-order sufficient condition for the search problem holds uniformly: $|\partial^2 W/\partial\theta^2| \ge \kappa > 0$ for all $(z,\chi,m,n) \in Z \times [0,1] \times K_L$.

Remark B.8. Assumption B.7 is required because C_2 in (B.3) involves value-function derivatives evaluated over K_L. Evaluating C_2 only at the prospective fixed point would be circular for the self-map argument; the assumption instead imposes the bounds as primitives and notes that they can be checked numerically at the solution.

Lemma B.9 (Invariance)**.** *Suppose Assumption B.7 holds and*

$$3C_2 < \varepsilon_p^2, \tag{B.2}$$

where

$$C_2 = \frac{(L_p^{\max})^2 \beta(1-d)}{\kappa}\left(2\|V^*\|\|\Pr{}'\|_\infty + \|\Pr\|_\infty\left\|\tfrac{\partial V_3^*}{\partial m}\right\|_\infty + \|1-\Pr\|_\infty\left\|\tfrac{\partial V_2^*}{\partial m}\right\|_\infty + \left\|\tfrac{\partial V_1^*}{\partial n}\right\|_\infty\right), \tag{B.3}$$

and $\varepsilon_p = \min(\underline{p}, 1-\bar{p}) > 0$ as defined in Assumption B.2. Then $\Psi : K_{L^*} \to K_{L^*}$ for $L^* = 2\varepsilon_p^{-2}/(1-3C_2\varepsilon_p^{-2}) < \infty$.

Proof. Bound on $|dm'/d\chi|$. Write $f(\chi) = \chi a(\chi)$ and $g(\chi) = \chi a(\chi) + (1-\chi)b(\chi)$, so $m' = f/g$ and, by the quotient rule, $dm'/d\chi = [f'g - fg']/g^2$.

Denominator. Since $p \ge \underline{p}$, we have $a, b \ge \underline{p}$, $g \ge \underline{p}$, and $g^2 \ge \underline{p}^2 \ge \varepsilon_p^2$.

Lipschitz bounds on a and b. The chain rule and $|\partial p/\partial\theta| \le L_p^{\max}$ give $|a'(\chi)| \le L_p^{\max}|d\hat{\theta}^*/d\chi|_{z=1}$ and similarly for $|b'(\chi)|$. To bound $|d\hat{\theta}^*/d\chi|$, apply the implicit function theorem to the first-order condition

$$F(\theta,\chi) \;:=\; -\nu'(\theta) + \frac{\partial p}{\partial\theta}\,\tilde{\beta}\left[\Pr(m(\chi))V_3^*(m(\chi)) + (1-\Pr(m(\chi)))V_2^*(m(\chi)) - V_1^*(n(\chi),z)\right] = 0.$$

Assumption B.7 ensures $|\partial F/\partial\theta| \ge \kappa > 0$ uniformly over K_L, so the theorem applies. Differentiating F with respect to χ, using $(m,n) \in K_L$ so that $|m'|, |n'| \le L$, and the

derivative bounds in Assumption B.7 gives $|\partial F/\partial\chi| \leq C_1 L$, where C_1 is the bracket in (B.3) multiplied by $L_p^{\max}\tilde{\beta}$. Hence $|d\hat{\theta}^*/d\chi| \leq C_1 L/\kappa$, and therefore $|a'|, |b'| \leq C_2 L$.

Collecting terms for m'. Since $|f|, |g| \leq 1$ and $a, b \in [\underline{p}, 1]$:

$$|f'| \leq 1 + C_2 L, \quad |g'| \leq 1 + 2C_2 L, \quad |f'g - fg'| \leq 2 + 3C_2 L.$$

Combining with the denominator bound, $|dm'/d\chi| \leq (2 + 3C_2 L)/\varepsilon_p^2$. Setting this $\leq L$ requires $L(1 - 3C_2/\varepsilon_p^2) \geq 2/\varepsilon_p^2$, which has a finite positive solution if and only if (B.2) holds. The choice $L^* = 2\varepsilon_p^{-2}/(1 - 3C_2\varepsilon_p^{-2})$ then satisfies $|dm'/d\chi| \leq L^*$.

Bound on $|dn'/d\chi|$. For $n'(\chi) = \chi(1 - a(\chi))/[\chi(1 - a(\chi)) + (1 - \chi)(1 - b(\chi))]$, write $\tilde{f} = \chi(1-a)$ and $\tilde{g} = \chi(1-a) + (1-\chi)(1-b)$. Since $p \leq \bar{p}$, we have $1 - a, 1 - b \geq 1 - \bar{p}$, so $\tilde{g} \geq 1 - \bar{p} \geq \varepsilon_p$ and $\tilde{g}^2 \geq \varepsilon_p^2$. The derivatives $|(1-a)'| = |a'| \leq C_2 L$ and $|(1-b)'| \leq C_2 L$ hold by the same Lipschitz argument. Hence $|\tilde{f}'| \leq 1 + C_2 L$, $|\tilde{g}'| \leq 1 + 2C_2 L$, and $|\tilde{f}'\tilde{g} - \tilde{f}\tilde{g}'| \leq 2 + 3C_2 L$, giving $|dn'/d\chi| \leq (2 + 3C_2 L)/\varepsilon_p^2 \leq L^*$. Therefore $\Psi(K_{L^*}) \subseteq K_{L^*}$. □

Remark B.10. Condition (B.2) jointly restricts $\underline{p}$, $\bar{p}$, $\alpha^{\max}$, β, d, and κ. Because C_2 depends on value function derivatives (Assumption B.7), the condition cannot be verified analytically, but it can be checked numerically. Economically, it requires the gap $\varepsilon_p = \min(\underline{p}, 1-\bar{p})$ to be sufficiently large relative to the sensitivity of optimal search effort to changes in beliefs.

Theorem B.11 (Existence)**.** *Under Assumptions B.1–B.7 and condition* (B.2)*, there exists a fixed point $(m^*, n^*) \in K_{L^*}$ of Ψ and a corresponding solution $(V_u, V_e^P, V_e^{NP}, m^*, n^*)$ to the full system in which the value functions solve the Bellman equations and the belief transition functions are consistent with optimal search behavior via Bayes' rule.*

Proof. K_{L^*} is compact and convex by construction, and $\Psi(K_{L^*}) \subseteq K_{L^*}$ by Lemma B.9. Continuity of Ψ on K_{L^*} follows from Lemma B.6 (for $(m, n) \mapsto \hat{\theta}^*$), continuity of $p(\cdot, z)$ (for $\hat{\theta}^* \mapsto (a, b)$), and smoothness of the Bayesian formulas in (B.1) with denominators bounded below by $\varepsilon_p > 0$ (for $(a, b) \mapsto (m', n')$). By Schauder's fixed point theorem, Ψ has a fixed point $(m^*, n^*) \in K_{L^*}$. Lemma B.5 then yields the unique $V^*(m^*, n^*) \in C_b(S)$ solving the Bellman equations at (m^*, n^*), completing the proof. □